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Exploring the interplay between infiltration dynamics and Critical Zone structures with multiscale geophysical imaging: A review

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ABSTRACT

Infiltration is the process of water entering into, and routings through, the subsurface. It has a profound impact on water flux in the Earth's Critical Zone (CZ), influencing phenomena such as overland flow, stream generation, plant available water, and groundwater recharge, as well as the associated CZ processes and functions of weathering, erosion, slope stability, and food production. However, reliable measurement of infiltration in structured geomedia (soils and regolith) with hidden subsurface features remain challenging. The increasing availability of geophysical techniques has helped them emerge as viable options for minimally invasive determination of infiltration at the pedon to hillslope or catchment scales. Geophysical methods replace the educated guesswork of interpreting field data from traditional infiltrometers with direct visualization and quantification. This paper provides a comprehensive literature review of various applications of geophysical techniques used to (1) visualize infiltration-relevant structures, (2) monitor non-uniform (or preferential) infiltration patterns, and (3) map soil moisture distribution after infiltration. It then proposes a new conceptualization (form-and-function dualism) for subsurface hydrology to link preferential infiltration to CZ structures, with the goal of formulating more realistic infiltration theories related to subsurface structural heterogeneity. The characterization of both subsurface structures (form) and infiltration dynamics (function) via geophysical imaging should help infiltration models, currently emphasizing macroscale behaviors rooted in the continuum concept, expand to accommodate the microscale nature of preferential flow. Integrating multiple geophysical approaches and developments into the new generation of geophysical techniques will further enhance our ability to see into the fabric of the subsurface and capture the dynamics of non-uniform infiltration through the hidden half of the CZ. This approach offers exciting opportunities for understanding the complex interactions between CZ structures and CZ processes.

1. Introduction

Infiltration is the process of water entering into the soil surface and its routing through the subsurface. It determines the partitioning of precipitation between the vadose zone and the residual accumulation and flow of water on the surface (Hillel, 1998). The rate of infiltration and the pathways it follows are controlled by the mode of water supply, antecedent soil wetness, surface microtopography, subsurface structures, soil properties, and land use/land cover (Rawls et al., 1983; Stephens, 1995). Infiltration strongly influences water fluxes through the Earth's Critical Zone (CZ), such as overland flow, stormwater runoff, plant available water, and groundwater recharge; as well as a variety of associated CZ processes, including soil erosion, stream generation, weathering of bedrock, and landslides (Bradford et al., 1987; Damiano

Abbreviations: AS, Amplitude similarity; CMP, Common midpoint; CRIM, Complex refractive index model; CS, Comprehensive similarity indicator; CT, Computed tomography; CZ, Critical Zone; DTW, Dynamic time warping; ECa, Electrical conductivity; EM, Electromagnetic; EMI, Electromagnetic induction; ERT, Electrical resistivity tomography; GCP, Ground control points; GPR, Ground-penetrating radar; RLPS, Rotary laser positioning system; SS, Structural similarity; TDR, Time-domain reflectometry; VES, Vertical electrical sounding

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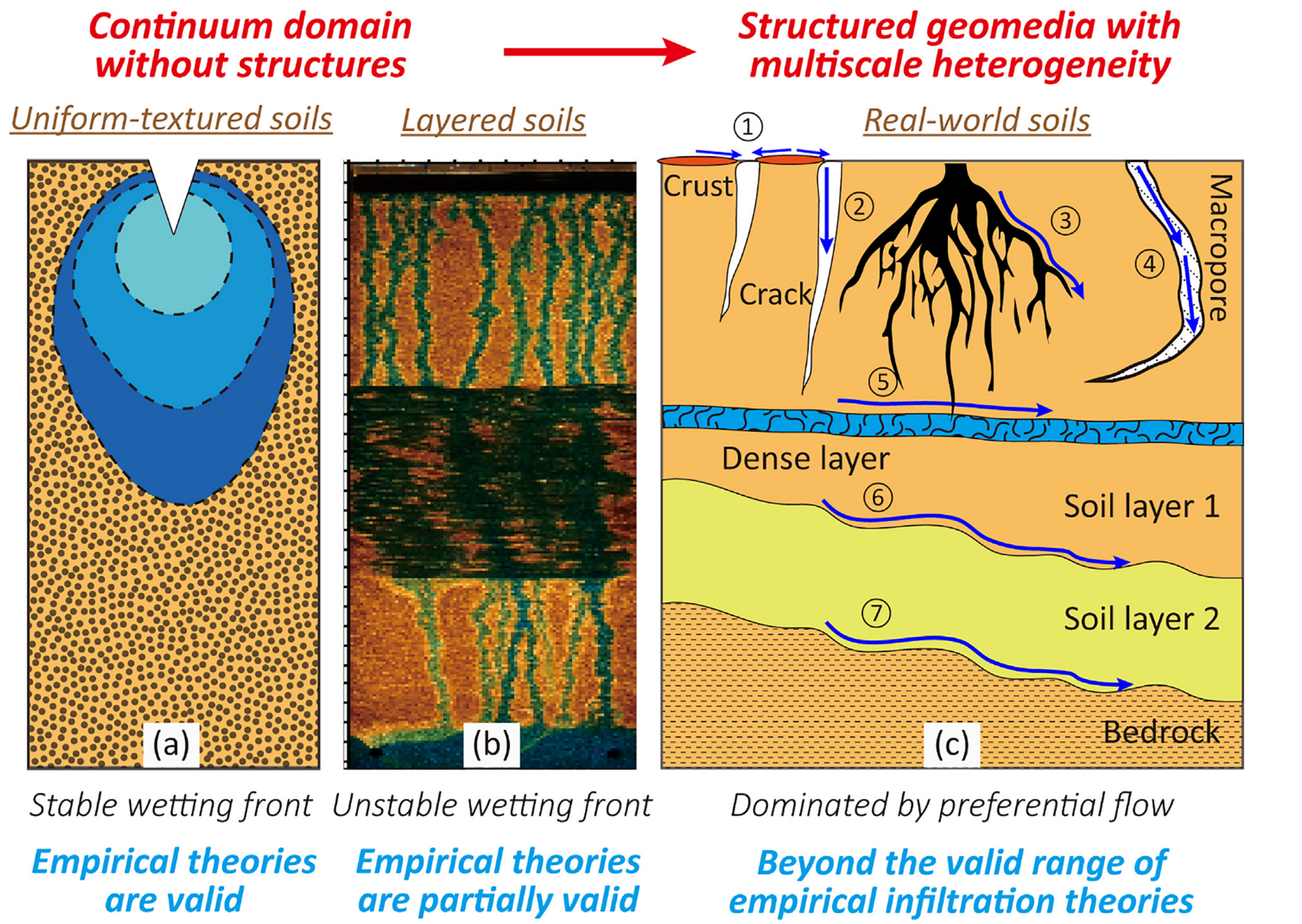

Fig. 1. A conceptual diagram of the validity of empirical theories when applied to different conditions, relevant as infiltration research shifts more attention to complex, real-world conditions. (a) Classical theories assume the uniform flow through a homogeneous medium via a stable wetting front. (b) In structured soils, unstable infiltration occurs, including finger flows in multi-layered, coarse-textured soils (adapted from Rezanezhad et al., 2006). (c) Various types of preferential flow coexist and dominate infiltration in real-world soils, which challenges the classical infiltration equations. The development of preferential flow is associated with different near-surface and subsurface structures depicted here, such as 1) surface sealing and crust, 2) soil cracks and fissures, 3) root channels, 4) macropores and soil pipes, 5) less-permeable layers, 6) interfaces between soil horizons, and 7) the soil-bedrock interface.

and Olivares, 2010; Lebedeva and Brantley, 2020). Knowledge of infiltration is, therefore, fundamental for efficient soil and water management, which impacts activities such as irrigation, food production, flood control, drought prevention, water quality assurance, and watershed management (Hillel, 1998; Huang et al., 2011).

However, our ability to describe and predict the dynamic processes of infiltration through the CZ, especially in soils and regolith with hidden subsurface structural features, is lagging behind our needs or is overly simplified (Lin, 2010). The field is struggling with a lack of realistic infiltration theories that address the complexity and heterogeneity in the subsurface of the CZ. Previous efforts have attempted to establish and model quantitative principles governing infiltration (e.g., Green and Ampt, 1911; Horton, 1941; Philip, 1969; Parlange, 1971). Empirical infiltration theories, such as the Green-Ampt equation and Phillip's model, have been confirmed effective to describe infiltration curves and soil matrix infiltration over time. However, these classical infiltration models are challenging or even impossible to predict specific infiltration pathways, because they assume uniform flow through a homogeneous medium (Fig. 1a), which is rarely true under natural conditions, where subsurface heterogeneity prevails and the flow of water through it is often irregular (Lin, 2010). Remarkably, these homogeneous domain theories are still the basis for most current models used to represent infiltration (e.g., Ogden et al., 2015; Fan et al., 2017). An infiltration theory that can predict the complex interactions between infiltration dynamics and subsurface structures remains to be formulated (Beven and Germann, 2013). Over the past several decades, increasing attention has been paid to the characterization of infiltration under more complex conditions, such as multi-layered soils (Fig. 1b; Huang et al., 2011), crust-topped soils (Eldridge et al., 2000), and swelling soils (Romkens and Prasad, 2006). However, it is still challenging, or even impossible, for current models to reproduce the infiltration processes observed in real-world soils (Assouline, 2013). Notably, preferential flow (i.e., nonequilibrium flow that bypasses large sections of the subsurface; Fig. 1c), has raised questions about the effectiveness of classical infiltration equations in structured geomedia (Guo and Lin, 2018).

In addition to theoretical complexity, reliable monitoring and quantification of infiltration at the field scale (e.g., from pedon to hillslope scales) with a high temporal and spatial resolution remain arduous. Traditional approaches to measuring infiltration include the use of drip infiltrometers, ring infiltrometers, and permeameters. However, factors, such as surface topography, subsurface heterogeneity, and flow instability, impair the efficacy of these approaches (Zhang et al., 2019). Besides, these methods assume a steady-state infiltration flux or saturated conditions, which are often difficult or impossible to achieve in the field, and the factors influencing the infiltrability before reaching the steady state are poorly understood (Smettem and Smith, 2002). As conducted at the ground surface, these methods provide limited information about infiltration dynamics through time, or infiltration patterns through space (Hillel, 1998; Lin, 2010). Thus, traditional infiltration measurements tend to ignore the underground complexity and infer the infiltration behaviors in the subsurface, which cannot reflect the complexity of non-uniform infiltration through the CZ. Even where individual measurements are accurate, the ability of point-scale measurements via infiltrometers and permeameters to represent field-scale hydrologic behaviors is limited. Although tracer experiments coupled with computed tomography (CT) has been used to precisely monitor infiltration through soil columns in the laboratory (e.g., Luo et al., 2010), small-scale infiltration patterns are challenging to incorporate into hydrological models that usually deal with larger spatial scales. The lack of a comprehensive theoretical foundation for water transport, mixing, storage, and release in structured geomedia adds uncertainty when designing appropriate strategies for field investigation (Binley et al., 2015).

Geophysical techniques have emerged as valuable means for the non-invasive (or minimally invasive) investigation of infiltration at a variety of scales of time and space. Time-lapse geophysical approaches, in particular, provide a new suite of tools to examine the temporal evolution and spatial distribution of non-uniform infiltration through the CZ (Parsekian et al., 2015). To date, ground-penetrating radar (GPR), electromagnetic induction (EMI), and electrical resistivity tomography (ERT) have been successfully used to study infiltration under field conditions. These technologies have allowed for the tracking of wetting fronts (e.g., Truss et al., 2007), the mapping of profile moisture distributions (e.g., Deiana et al., 2008), and the delineation of preferential flow pathways (e.g., Guo et al., 2014). This detailed information supplements the educated guesswork of interpreting infiltration processes via traditional methods with direct visualization and quantification. Specific advantages of using geophysical imaging to study infiltration include their ability to:

- reveal the true overall infiltration patterns and flow pathways, rather than assuming uniform flow;
- cover a range of spatial scales (from several cubic meters to 10^6 cubic meters) instead of point measurements;
- permit repeated, non-destructive surveys to capture transient dynamics, instead of relying on steady-state conditions; and
- link infiltration dynamics to subsurface structures. The establishment of the relationship between flow dynamics and structural heterogeneity is a long-standing need in multiphase flow research (Armstrong et al., 2016) and the emerging interdisciplinary science of hydropedology, as well as the CZ science (Guo and Lin, 2016).

The objective of this paper is three-fold: (1) to review the state-of-knowledge of infiltration field measurements by geophysical imaging; (2) to propose a new conceptualization of subsurface hydrology that describes the interaction between non-uniform infiltration and structured geomedia in the CZ; and (3) to provide predictions and recommendations to improve the future application of geophysical methods in monitoring and quantifying infiltration. Given the increasing use of geophysical techniques in CZ studies, this paper aims to stimulate and guide the ongoing efforts to understand the complex interactions between CZ structures and CZ processes.

2. Principles of geophysical techniques

The most frequently used geophysical methods of measuring infiltration in the field, especially for near-surface investigation (within a few meters deep), include GPR, ERT, and EMI. This section introduces the working principles of these techniques and the criteria of selecting the proper method(s) for a given application.

2.1. GPR

The GPR method captures changes in electromagnetic (EM) properties within the shallow subsurface, especially its dielectric permittivity, which is a general measurement of how well EM energy is transmitted through a medium (Huisman et al., 2003). High-frequency EM energy (usually ranging from 10 to 2000 MHz) is generated by a transmitting antenna and propagates into the ground as waves. When radar waves pass across interfaces between media with different dielectric permittivities, a portion of their energy will be reflected back, where it may be detected by a receiving antenna, while the remainder continues to propagate deeper until it is entirely attenuated. The travel time, amplitude, shape, and polarity of a reflection can be used to determine the location, size, and dielectric permittivity of the subsurface object or interface that caused it. Because most geologic materials have dielectric constants (i.e., relative permittivities) in the range of 3 to 30, whereas the dielectric constant of water is 81, even small changes in soil moisture will significantly alter radar reflection wave velocities and reflection amplitudes (Huisman et al., 2003).

2.2. ERT

The use of ERT is an active source method that uses a low-frequency electrical current (usually $<$ 50 Hz) to measure electrical resistivity distributions in the subsurface. An ERT survey is carried out by placing an array of electrodes on the ground at a specific interval. The number and spacing of electrodes determine the detection resolution and the penetration depth (Binley and Kemna, 2005). Direct current is injected into the ground between one pair of electrodes and the system measures the difference in voltage between another pair. This process is repeated with different electrodes serving as transmitter and receiver. Apparent resistivity can be calculated from the current and voltage values by assuming a homogeneous half-space without geological structures (Michot et al., 2003). After an inversion of the apparent resistivity, the true spatial distribution of resistivity can be determined in a 2D cross-section. Electrical resistivity varies with porosity (bulk density), clay content, temperature, soil moisture, the conductivity of the pore water, and salinity (Parsekian et al., 2015), which makes ERT a possible tool for imaging both subsurface structures and hydrogeological processes (Chambers et al., 2014).

2.3. EMI

The EMI method measures the bulk or apparent electrical conductivity (ECa; a measure of charge mobility due to the application of an electric field) of earthen materials (Doolittle et al., 2012). An EMI system transmits low-frequency circular eddy-current loops into the subsurface, and the magnitude of these loops is proportional to the subsurface electrical conductivity (Corwin and Lesch, 2005). Each current loop induces a secondary electromagnetic field, which is detected by a receiver coil. The amplitude and phase of the secondary field reflect subsurface properties, such as soil texture, clay content, mineralogy, soil moisture, soil temperature, and salinity (Corwin and Lesch, 2005). Unlike GPR or ERT, an EMI system can be operated without the need to establish a coupling with the ground, which allows rapid acquisitions from airborne or towed platforms. Once other variables are controlled for, EMI can be used to measure soil water content and subsurface flow (e.g., Robinson et al., 2012).

2.4. Selection of geophysical methods

Choosing the best geophysical tool(s) for a specific application requires careful consideration of the scale and the features of interest. Fig. 2 summarizes the tradeoffs of these geophysical methods, as they relate to the depth of investigation, the horizontal area of investigation, the spatial resolution of flow-relevant structure identification, and the interval timing during time-lapse surveys.

In general, GPR has a higher spatial resolution and data collection rate, but a minimal detection depth that is influence by antenna frequency and local conditions; ERT has a greater depth of investigation,

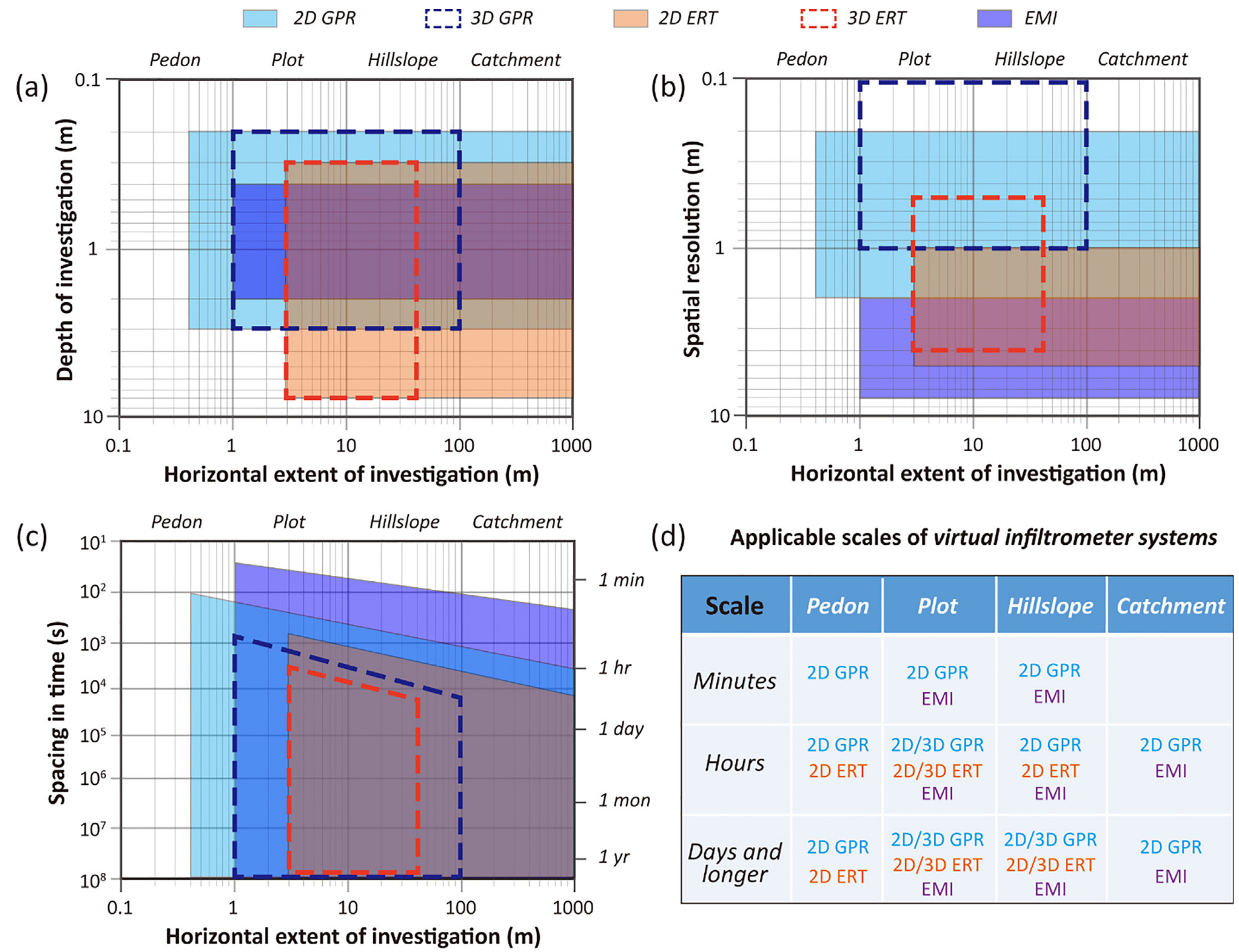

Fig. 2. Depictions of the tradeoffs among investigation depth, horizontal extent of investigation, spatial resolution, and time interval in time-lapse surveys of GPR, ERT, and EMI (a, b, and c, respectively), based on the authors' experience and data synthesized from Robinson et al. (2008), Binley et al. (2015), and Walvoord and Kurylyk (2016). (d) Possible combinations of multiple geophysical techniques to determine infiltration across time and space scales.

but also a confined spatial coverage and a lower rate of data collection; and EMI can be used to survey a large area, but the spatial resolution and the depth of investigation are limited (Fig. 2). Besides, EMI usually produces results as a depth-weighted average and is susceptible to temperature variations. All these techniques can be used in a time-lapse manner. Because the electrodes remain in the same location during data collection, time-lapse ERT images from the same survey transect have identical geometry. However, repeating GPR and EMI surveys with identical geometry is challenging in the field. Sometimes, a positioning system is attached to the antenna of GPR and EMI to allow real-time recording of coordinates and have better control of the survey area. The use of 3D GPR and ERT surveys can increase their spatial resolution than 2D survey lines, but such data takes longer to collect, especially across large areas or when monitoring the transient dynamics of infiltration (Fig. 2). Combinations of geophysical methods can provide an effective suite of tools to observe and quantify infiltration through the CZ (see Section 5.1).

3. Examples of geophysical characterization and observation of infiltration

3.1. Visualizing infiltration-relevant structures in subsurface

To date, GPR has been used to delineate various infiltration-relevant structures in the CZ, such as soil macropores (e.g., Gormally et al., 2011), animal burrows (e.g., Kinlaw and Grasmueck, 2012), coarse roots (e.g., Guo et al., 2015), soil horizonation (e.g., Han et al., 2016), the bedding orientation of bedrock (e.g., Nyquist et al., 2018), fractures in weathered bedrock (e.g., Guo et al., 2019), and water-restricting layers (e.g., Doolittle et al., 2012). The ERT method has successfully imaged soil pipes (Leslie and Heinse, 2013), soil-bedrock interfaces (Leucci, 2010), permafrost layers (Hilbich, 2010), and mire base layers (Sass et al., 2010). By adopting multi-receiver EMI sensors or multi-layer inversion modeling algorithms, some studies have applied EMI for the detection and locating of subsurface structures, such as soil horizons, drainage ditches, and paleochannels (De Smedt et al., 2013a; Saey et al., 2012). This detailed information on subsurface geometry can help understand the interaction between the non-uniform infiltration in the CZ and subsurface structures. Examples of geophysical characterization of soil macropores, coarse roots, water-restrictive layers, and drainage pipes are discussed below.

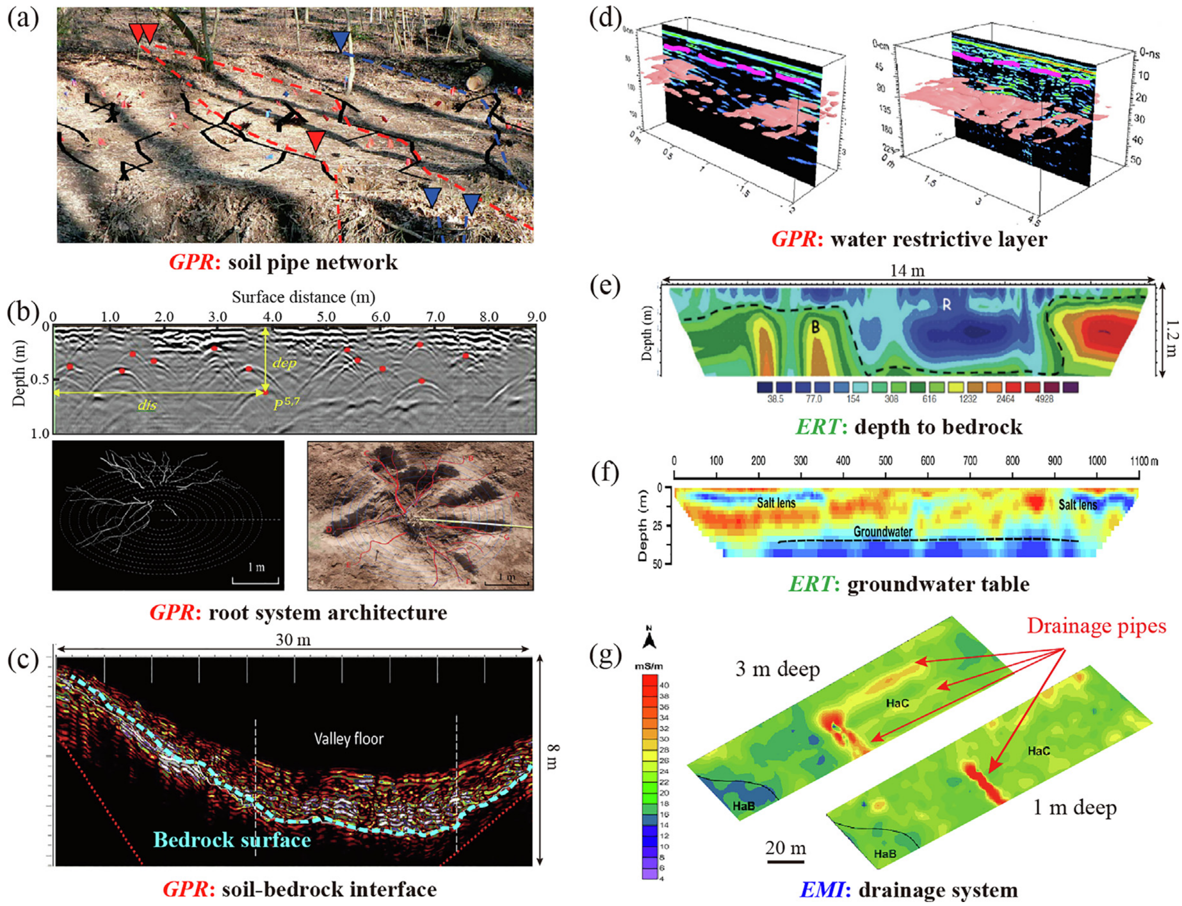

Fig. 3. Visualizations of infiltration-relevant subsurface structures. (a) Locations of soil pipes mapped in a riparian zone (adapted from Gormally et al., 2011). The GPR-derived macropores (solid curves) were confirmed by preferential dye conduction from input to the stream (dashed curves). (b) An image of coarse root distribution in sandy soils (adapted from Wu et al., 2014). Lower left: 3D root system architecture reconstructed by connecting root points identified in GPR images (indicated by the red dots in the upper panel). Lower right: comparison between excavation and the reconstruction model (red curves). (c) A map of soil depth to bedrock in a forested catchment (adapted from Doolittle et al., 2012). The dashed curve indicates the interpreted soil-bedrock interface. (d) A map of water-restricting layers (indicated by the pink isosurface) in two contrasting soils (adapted from Zhang et al., 2014). (e) The 2D ERT model for the top 1.2 m of the subsurface. "B" indicates base bedrock with higher resistivity values; "R" indicates weathered soil with lower resistivity values (adapted from Leucci, 2010). (f) Map of the depth to groundwater table and the continuity of the salt lens in a 1.1-km-long ERT profile (adapted, with permission, from Jayawickreme et al., 2011). (g) A map of subsurface drainage pipes, which generated extremely high ECa values in EMI images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1.1. *Macropore structures*

Soil macropores are recognized as primary pathways for the preferential routing of infiltration water (Beven and Germann, 2013). In a riparian wetland, Gormally et al. (2011) tested the feasibility of a 900-MHz GPR system to map the 3D structure of lateral macropore (diameter > 3 cm) networks at depths of 15–65 cm. The macropore locations detected by GPR were ground-truthed by soil coring, and this test suggested an accuracy of 92% for GPR-derived macropore locations. Tracer dye transmission then corroborated the macropore network morphology predictions suggested by the GPR results (Fig. 3a). Another study demonstrated the efficacy of GPR (with 100 and 200 MHz antennae) in locating two soil pipes (one with a diameter of ~15 cm at a depth of 180 cm and the other with a diameter of ~28 cm at a depth of 300 cm) and mapping their connectivity in blanket peat on a hillslope (Holden, 2004).

Leslie and Heinse (2013) used pseudo-3D ERT to characterize the spatial distribution of soil pipes within a forested hillslope. ERT surveys were conducted on three ~6 × 6 m^2 plots underlain by a fragipan. Excavation after the ERT survey showed that soil pipes at this particular site were mostly empty, resulting in elongated high-resistivity features on the ERT profiles. However, the high resistive fragipan layer impaired the efficiency of ERT to delineate soil pipes, and only a small portion of soil pipes (with a diameter ~10 cm and at a depth of 30 to 40 cm) were observable in ERT images. Also, there was no clear relationship between the diameter of a soil pipe and the detection frequency of soil pipes by ERT. The authors argued that the enhanced electrical resistivity contrast achievable by releasing chemical tracers, or by conducting ERT surveys after strong precipitation when the soil pipes are transporting water, might aid in the detection of soil pipes (Leslie and Heinse, 2013). More recently, high-resolution ERT profiles collected with a 0.5-m electrodes interval were used to map soil pipes and the continuity of the pipe network down to 4 m deep with 3D analysis (Bovi et al., 2020). The authors suggested resistive zones of values greater than 4000 O.m as an effective threshold to detect soil pipes in ERT profiles (Bovi et al., 2020). Another study indicated that soil cracking increased bulk soil electrical resistivity significantly, and, thus, in situ electrical resistivity measurements by 3D ERT held the potential to map soil cracks at the meter scale (Ackerson et al., 2014).

3.1.2. *Coarse roots*

Root channels are important pathways that route infiltrating water through the soil (Guo et al., 2020; Li et al., 2009). In the arid and semi-arid shrublands of northern China, where the sandy soil provides an ideal condition for GPR-based root detection and quantification, a series of studies have been conducted to estimate root diameter and biomass (Guo et al., 2015), to reconstruct the architecture of coarse root systems (Wu et al., 2014; Fig. 3b), and to map the density distribution of roots with soil depth (Cui et al., 2020). The potential of GPR to map coarse roots has also been validated in temperate forests (e.g., Molon et al., 2017), coastal forests (e.g., Butnor et al., 2001; Ohashi et al., 2019), agroecosystems (e.g., Borden et al., 2017), and even in urban environments (e.g., Nichols et al., 2017). These studies indicated that GPR is most effective in detecting lateral roots with a diameter larger than 2 cm in dry sandy soils, especially when the cross angle between a root branch and the survey line is larger than 45° (Hirano et al., 2012; Guo et al., 2015) but less effective for vertical growing roots (e.g., taproots) and in soils with a high organic content or a high water content (Butnor et al., 2001, 2016).

3.1.3. *Water-restrictive layers*

A transient water table can perch above a water-restricting layer and trigger continuous lateral preferential flow downslope (Lin, 2010). In the permafrost zone on the Tibetan Plateau, Ma et al. (2015) employed a 250-MHz GPR system to map seasonal frost depth under different topographical and surface conditions at the beginning of the freezing and thawing periods. They found that topography, vegetation, and distance to the stream significantly influenced the spatial distribution of seasonal frost depth at the hillslope scale. Doolittle et al. (2012) used 200- and 400-MHz GPR systems to map the soil depth to bedrock across a 7.9-ha forested catchment. At the soil-bedrock interface, GPR reflections transitioned from high-amplitude (due to the relative structural heterogeneity of contrasting alluvium, colluvium, and saprock) to low-amplitude (due to the relative structural homogeneity of the unweathered bedrock; Fig. 3c). The comparison between GPR interpretation and *in situ* observation showed a reasonable accuracy. In the same catchment, Zhang et al. (2014) used a 400-MHz GPR instrument to detect the micro-topography of water-restricting layers in different soil types (Fig. 3d) and proposed the mechanistic link between seasonal preferential flow dynamics and water accumulation above the water-restricting layer.

Leucci (2010) used the ERT method to visualize the soil-bedrock interface of a former quarry. The ERT section showed a low-resistivity layer (interpreted as soil cover) that overlaid high-resistivity bedrock (Fig. 3e), which matched GPR results obtained at the same site. Moreover, ERT provided extra information that allowed the researchers to distinguish the root zone and the old pipe networks that were indistinguishable on GPR profiles (Leucci, 2010). Sass et al. (2010) combined conventional investigation techniques (telescope rods and drilling) with GPR and ERT to explore the internal structure of alpine mires. They found that the 100- and 200-MHz GPR antennas could detect the base layer of the mires with high accuracy. However, the internal interfaces of the mires were less apparent in the GPR images, which might be explained by the relatively weak contrasts in dielectric constants at the interfaces between peat types, or across decomposition stages. ERT provided additional information about the mire base layer and the deeper subsurface that was inaccessible to GPR (Sass et al., 2010). In addition, Jayawickreme et al. (2011) successfully used ERT to locate groundwater depth and to map the extension of the salt lens in a woodland–agriculture–woodland agroecosystem (Fig. 3f). The depth of the high-conductivity horizon might indicate the presence of a water-restricting layer, but the absence of the salt lens under the crop field suggested that irrigation enhanced deep infiltration and promoted salt leaching from the vadose zone and into the groundwater (Jayawickreme et al., 2011).

3.1.4. *Drainage systems*

In agricultural and urban environments, man-made drainage systems significantly influence water infiltration and routing in the subsurface. We used the Dualem-2 EMI meter to map subsurface drainage pipes over a wastewater spray field. EC_a measurements were strongly affected by a buried drainage pipe, which partially crosses the lower-center portion of the plots at 1 m depth (Fig. 3g). Two additional lateral pipes appeared at 3 m depth, which extended in a northeasterly direction away from the known buried pipe (Fig. 3g). There are a number of successful applications of GPR to map agricultural pipes (e.g., Allred et al., 2018) and drainage systems in cities (e.g., Zhao and Al-Qadi, 2017).

3.2. *Monitoring infiltration processes*

Because the dielectric permittivity, electrical conductivity, and electrical resistivity (i.e., the geophysical properties measured by GPR, EMI, and ERT, respectively) are all sensitive to the change in soil moisture, a small variation in profile moisture distribution can result in different geophysical images. Therefore, time-lapse measurements have become increasingly popular as a multidimensional suite of investigation methods to look into the infiltration dynamics in the subsurface environment, such as tracing the displacement of the wetting front and mapping subsurface flow pathways of the infiltrated water.

3.2.1. *Wetting front displacement*

Dietrich et al. (2014) combined time-lapse ERT with successive tensiometer and time-domain reflectometry (TDR) measurements to trace the movement of wetting fronts after water was sprayed over a petrocalcic Paleudalfs in Argentina (Fig. 4a). Their results demonstrated that the petrocalcic horizon of the study soil impeded the downward movement of the wetting front and triggered lateral flux above this water-restricting layer, whereas the less-cemented zones in the petrocalcic horizon favored the development of vertical preferential flow (Dietrich et al., 2014). Chambers et al. (2014) applied 28 time-lapse 3D ERT measurements over a railway embankment to monitor soil moisture dynamics over 17 months. The resistivity-moisture relationship was established in the laboratory using soil from the study site. After the effects of temperature changes on resistivity were corrected, the collected 3D ERT data were translated into soil moisture. The observed seasonal wetting fronts on these 3D ERT profiles correlated closely with the recorded effective rainfall. In a maze field, a recent study employed high-resolution ERT profiles collected with a 0.15-m electrodes spacing to visualize the displacement of wetting zones and root water uptake effects over time in an irrigation circle (De Carlo et al., 2020).

Truss et al. (2007) conducted time-lapse GPR surveys with a 3-min interval. They collected measurements before, during, and after a rainfall event to study the drainage process in the Miami Limestone in Florida, U.S.A. Their results indicated that repeated GPR profiles could provide assessments of the location of preferential infiltration zones, the geometries of preferential flowpaths, and wetting front displacement. Propagation rates (both vertical and horizontal) could also be estimated by comparing the different stages of the infiltration (Truss et al., 2007). Under laboratory conditions, time-lapse GPR has been employed to capture the transient dynamics of wetting front movement. For example, Mangel et al. (2015) developed an automated GPR-data collection system to monitor the non-uniform propagation of a wetting front through the soil column (refilled river sand) with a high spatio-temporal resolution (Fig. 4b). More recently, Mangel et al. (2020) coupled this automated high-speed GPR data acquisition platform with reflection tomography algorithm to depict the transient 2D soil moisture distribution and the edges of the wetting front after infiltration, where preferential flow response obscured the generation of the coherent wetting front reflection in GPR images. However, in structured soils, the signal of the wetting front is likely to be difficult to spot with GPR, as it is obscured by subsurface heterogeneity. To obtain a robust measure of the changes in time-lapse GPR images with low signal-to-noise ratios, Allroggen and Tronicke (2016) introduced a methodology from modern image processing to calculate the structural similarity attribute. Later, Jackisch et al. (2017) applied this method to trace

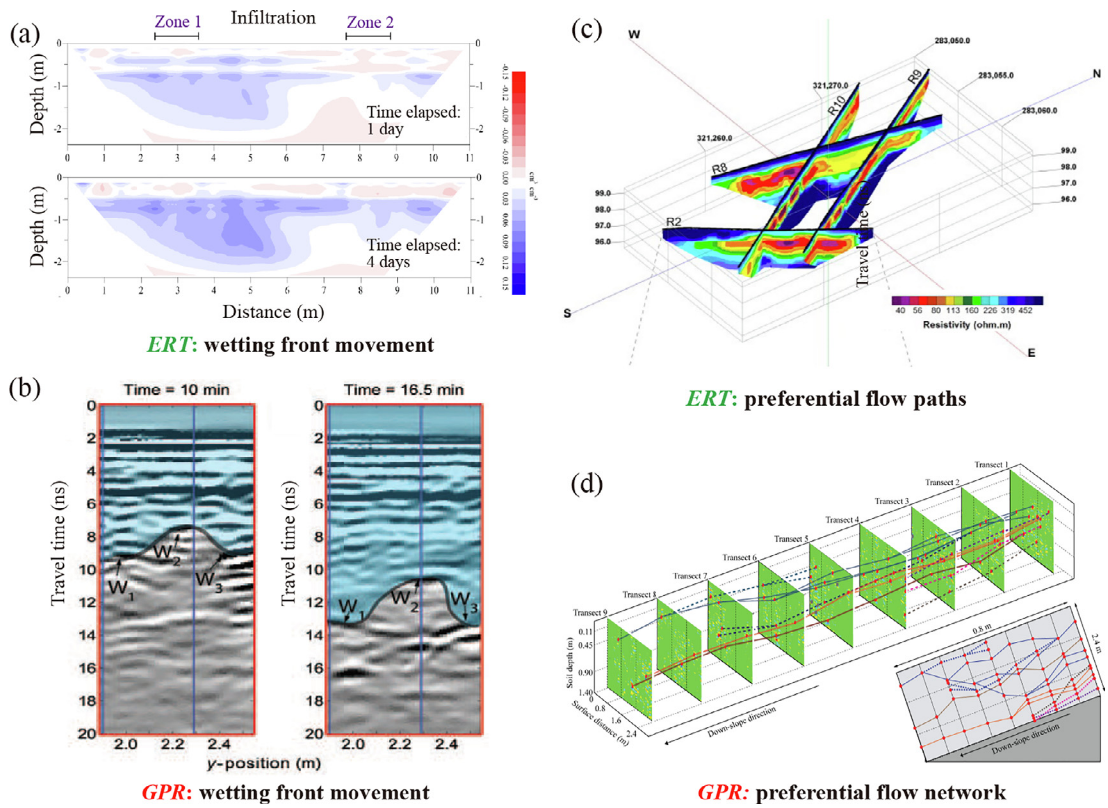

Fig. 4. Examples of monitoring wetting front displacement and mapping preferential flowpaths with geophysical tools. (a) Tracing the non-uniform movement of the wetting front by time-lapse ERT (adapted from Dietrich et al., 2014). Blue shading indicates wetter soil. (b) Images of the non-uniform movement of a wetting front (W1, W2, and W3) by time-lapse GPR (adapted from Mangel et al., 2015). (c) A 3D diagram of major subsurface preferential flow pathways associated with a contaminant flume by ERT (adapted from Donohue et al., 2015). Warm colors indicate lower resistivity values and, thus, higher contaminant concentrations. (d) A reconstruction of subsurface lateral preferential flow network on a forested hillslope by time-lapse GPR coupled with infiltration experiments (adapted, with permission, from Guo et al., 2014). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

wetting front movement after irrigation and natural rainfall events by locating the areas with great time shifts in time-lapse GPR images.

3.2.2. Subsurface preferential flowpaths

Donohue et al. (2015) used ERT to investigate contaminant transport in on-site wastewater treatment systems over a low-productivity aquifer. They found that the spatial distribution of chemical and microbiological contaminant levels inferred from ERT measurements agreed with groundwater samples. Their ERT profiles registered an area (14 m wide and 1.5 m thick) of gross groundwater contamination in the vicinity of the percolation area as a zone of lower resistivity than its surroundings (Fig. 4c). Groundwater samples and ERT results confirmed that the main body of the effluent plume traveled deeper into the subsoil. The interface between weathered shale and fresh bedrock might provide a preferential route for wastewater transport, where hydraulic conductivities appeared to be higher. A similar study applied time-lapse 3D ERT to monitor precipitation and saline tracer transport in a 72-m^2 plot in ditch-drained agroecosystems after a rainstorm (Robinson et al., 2020). Vertical preferential flowpaths of the tracer plume were identified in the shallow argillic horizon, followed by steady lateral transport towards the drainage ditch in the deeper sandy horizon (Robinson et al., 2020). Moreover, the EMI method has also been used to identify preferential flow paths of the leachate plume associated with a municipal landfill (Triantafilis et al., 2013).

On planar hillslopes covered by thin soils in a forested catchment, time-lapse GPR revealed strong vertical preferential flow through the soils and the fractured bedrock, but limited development of downslope lateral preferential flow (Doolittle et al., 2012; Nyquist et al., 2018). By contrast, the deep soils located in concave hillslopes in the same catchment were found to harbor extensive lateral preferential flow through networks of macropores and above a low-permeability layer (Guo et al., 2014; Fig. 4d). Moreover, preferential flow along the fractured shale bedrock could direct water from planar and convex hillslopes to concave hillslopes, where the infiltrating water can travel downslope (Guo et al., 2019). Most recently, Di Prima et al. (2020) combined time-lapse GPR with automatized single-ring infiltrometers to detect the pathways of infiltrated water in a stormwater infiltration basin in France. They detected two major types of preferential flow, i.e., that coarse gravels acted as capillary barriers to converge infiltrated water into narrow pathways as funnel flow and macropore flow along plant roots. The preferential infiltration pathways were confirmed by direct observation of dyed patterns. These studies demonstrated the effectiveness of GPR in monitoring fast subsurface flow processes on a scale of minutes to hours. Another example of integrating GPR surveys and dye staining to identify different types of preferential flow is Haarder et al. (2011). They compared the abilities of time-lapse GPR and a dye tracer injection to map unstable wetting fronts and preferential flow. They infiltrated 900 L of Brilliant Blue dyed water over a 3 m by 3 m area that was underlain by relatively homogeneous and undisturbed sandy alluvial sediments. They found significant changes in the GPR reflection amplitude and reflection delay within the dye-stained area. Below the extent of the dye-stained area, GPR data suggested the existence of bypass flow, fingering flow, and lateral flow. Their results indicated that high-resolution GPR is a suitable method to map complex flow phenomena in the vadose zone, especially deep, unsaturated flow that is particularly challenging to detect by dye tracer release (Haarder et al., 2011).

Six EMI measurements have been conducted over a 19.5-ha agricultural field at different times of year for two years, the results of which suggested that the areas with the greatest seasonal variations in

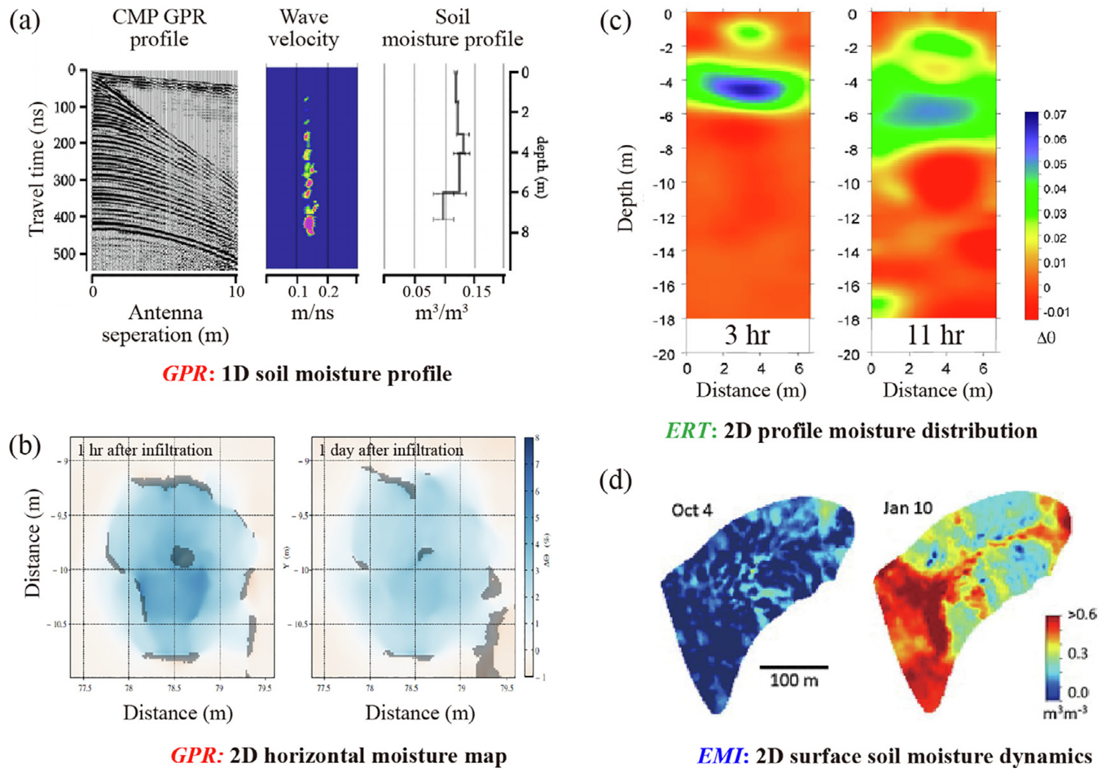

Fig. 5. A collection of maps and profiles of subsurface and near-surface soil moisture distribution, as derived from geophysical tools. (a) Measurements of the 1D vertical soil moisture profile by common midpoint (CMP) GPR. Left: CMP GPR image; middle: semblance analysis to obtain the most likely wave velocity at a certain depth; right: calculation of soil moisture levels from wave velocity using Topp's equation (adapted, with permission, from Comas et al., 2014). (b) Maps of subsurface horizontal moisture distributions at different times after infiltration by time-lapse, common-offset GPR. The color bar indicates changes in soil moisture, with blue reflecting an increase in soil moisture (adapted, with permission, from Allroggen et al., 2015). (c) Conversion resistivity in the 2D vertical soil moisture profile using the calibrated Archie's law (adapted, with permission, from Deiana et al., 2008). The color bar indicates changes in soil moisture, with blue indicating an increase in soil moisture. (d) Seasonal changes in the distribution of near-surface soil moisture, as mapped by time-lapse EMI surveys (adapted, with permission, from Robinson et al., 2012). The color bar indicates the average near-surface soil moisture. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ECa were more hydrologically active and were located closer to simulated lateral preferential flow pathways (Zhu et al., 2010a, b). As most soil properties that control ECa (e.g., texture, organic matter, and depth to bedrock) should remain largely unchanged over the survey period, the differences between repeated EMI scans must be related to changes in soil moisture (Zhu et al., 2010a, b). Based on the ECa variations between the dry and wet seasons, Doolittle et al. (2012) identified several lateral preferential flowpaths connecting ridge tops to the valley floor in a 7.9-ha catchment.

3.3. Mapping moisture distribution

A common approach to inferring soil moisture values from GPR scans is through common midpoint (CMP) measurements, where the transmitter and the receiver are separated by increasingly large distances around a static midpoint. For example, Comas et al. (2014) conducted CMP GPR surveys in a peatland in western Wales, U.K., to obtain a 1D vertical velocity model using semblance analysis (Parsekian et al., 2010), which identified the most likely GPR wave velocity from a spectrum of potential velocities for each depth (Fig. 5a). The measured wave velocity can be related to dielectric permittivity and, by extension, to soil moisture through empirical relations such as Topp's equation (Klotzsche et al., 2018). Koyama et al. (2017) improved the time efficiency and accuracy of CMP measurements by implementing a semi-automated data acquisition technique. They further enhanced signal processing of the CMP data to obtain reliable information on dielectric permittivity and the depth of the reflecting soil layers, which led to the accurate estimation of vertical soil moisture profiles.

In addition to 1D vertical soil moisture profiles, time-lapse common-offset GPR surveys (i.e., with a fixed spacing between transmitter and receiver) have been used to map 2D horizontal soil moisture in the subsurface. Using a specific horizontal reflector as a reference in time-lapse GPR images allows the estimation of the GPR wave velocity change caused by infiltration and wetting, which can be used to quantify the changes in soil moisture through the use of the complex refractive index model (CRIM). Assuming the validity of the CRIM, the change in soil moisture is linearly related to the change in GPR wave velocity (Robinson et al., 2003). If multiple distinct horizontal reflectors can be identified in the GPR images, the travel time differences can be calculated for each of the reflectors, and the dynamics of soil moisture at different depths can be monitored. For example, Allroggen et al. (2015) developed a metal-free platform to assist in the collection time-lapse GPR data with high re-positioning accuracy. The change in the travel time of GPR waves to the same horizon after infiltration was used to calculate the slowness in GPR wave velocity due to infiltrating water. They established a linear relationship between the change in soil moisture and the slowness in wave velocity and generated soil moisture maps (Fig. 5b). Steelman et al. (2012) collected both the CMP data and GPR reflection profiling data over a 26-month period to quantify soil moisture distributions and dynamics at an agricultural site in Canada. They obtained the changes in travel time between four stratigraphic reflections on different experiment dates, and the depths of the reflecting interfaces were constrained by the CMP data, based on which the interval-travel time data were converted into high-resolution soil moisture data. The GPR-derived soil moisture time series were able to reflect the transient infiltration pulses and short-duration wetting/

drying and seasonal freezing/thawing processes.

Schmelzbach et al. (2012) developed a reflection amplitude inversion method to resolve subsurface dielectric permittivity and related soil moisture distribution with much-improved resolution. When applied to constant-offset radar data collected by a 100-MHz GPR system over a shallow sedimentary aquifer, their method was able to map the distribution of soil moisture of the shallow saturated zone (3–7 m depth) with decimeter resolution (Schmelzbach et al., 2012). In addition, hyperbolic reflections existing throughout the 2D common-offset GPR images can be selected to estimate the mean wave velocity to the depth of the reflector. Most recently, Liu et al. (2019) and Illawathure et al. (2020) adopted an automatic hyperbola detection algorithm to identify reflections generated by point reflectors (e.g., coarse roots) in GPR images and determine the wave velocity, which then was used to calculate the average soil moisture from the surface to the depth of each identified point reflector. Liu et al. (2020) further enhanced the method by incorporating 3D interpolation to produce the 2D and 3D visualizations of root zone soil moisture (0 to ~1 m deep) in two field plots (30 m by 30 m). Both the absolute values and spatial distributions of soil moisture derived by the GPR method matched reasonably those obtained by augering. These efforts extend the application of ground-coupled GPR from mapping surface soil moisture to the spatial distribution of subsurface soil moisture. Dou et al. (2017) proposed a method for real-time hyperbola recognition and fitting in GPR images, which was verified on both synthetic and field-collected GPR data. They found that the proposed method is suitable for real-time on-site applications due to its low computation cost and high accuracy. Whether this method can be further improved to map soil moisture distribution by GPR in a real-time manner is worthy of follow-up studies.

A 2D electrical resistivity profile can also be converted to a 2D vertical soil moisture profile (Samouelian et al., 2005). Prior to field geophysical surveys, preliminary calibration of the volumetric soil water content related to electrical resistivity was required, in order to obtain the site-specific parameters for Archie's law (Michot et al., 2003). Variations in the 2D vertical soil moisture profile could then be derived from the electrical resistivity data acquired at different times. For example, in an experimental site characterized by Quaternary sand and gravel sediments in Italy, Deiana et al. (2008) applied time-lapse ERT in a 2D cross-hole configuration to monitor the infiltration process after ~20 m^3 of fresh water was released into the vadose zone. They used core data to develop an Archie's law relationship between soil bulk resistivity and soil moisture along a gradient of saturation levels under laboratory conditions. Finally, the 2D resistivity images were converted to 2D vertical soil moisture maps using this calibrated Archie's law relationship (Fig. 5c). They found that the movement of the wetting front and the water content dynamics at different depths within the vadose zone (0–20 m deep) could be accurately monitored by time-lapse borehole ERT. Based on the differences in electrical resistivity, high-resolution ERT with a 1-m spacing between electrodes was used successfully to map the distribution of freshwater lens, capillary fringe, and the salinized coastal aquifer at the meter scale (Greggio et al., 2018). Furthermore, Boaga et al. (2013) used 3D ERT surveys to monitor 3D soil moisture dynamics of a soil cube (0.9 m wide, 0.9 m long, and 1.2 m deep) in an apple orchard after irrigation, which revealed the displacement of the wetting front beneath the irrigation drippers. Future studies should test 3D ERT in mapping soil moisture at larger spatial scales.

The EMI method has been applied to investigate near-surface soil moisture dynamics over a large area. Robinson et al. (2009) investigated the spatiotemporal variation of soil water redistribution based on a sequence of EMI surveys on a 4-ha deltaic field before and after a strong rainfall. In the non-saline soils at their study site, soil moisture was found to be the major contributor to the temporal changes in ECa. The comparison of the spatial distribution of ECa mapped during eight survey dates delineated the zones of water depletion and accumulation (Robinson et al., 2009). Similar time-lapse EMI surveys were performed in a semi-arid oak savanna over a six month period to map near-surface soil moisture patterns (Robinson et al., 2012). They reported that, within this small catchment (~4 ha), soil moisture responses to precipitation infiltration varied spatially, mostly due to differences in topography, vegetation, and soil texture (Fig. 5d).

Despite the high potential of geophysical methods in mapping soil moisture distribution, their successful application is always site-specific. For example, GPR is less effective in areas with relatively high electrical conductivity where the attenuation of the EM wave is more considerable (Klotzsche et al., 2018). Field sites with rough terrain or a thick litter layer also impact the accurate estimation of wave velocity and, thus, soil moisture by GPR. EMI produces soil moisture maps as a depth-weighted average, which could not reflect soil moisture variation with soil depth (e.g., Robinson et al., 2012). The limited spatial resolution of ERT and EMI may not be able to capture localized soil moisture variations (Liu et al., 2020). Another limitation is the uncertainty in the inversion of geophysical data to soil moisture due to the indirect nature of geophysical methods. For example, an Archie's law relationship between soil resistivity and soil moisture is required to convert ERT measurements to soil moisture. However, due to the heterogeneity of real-world soils, even at a small scale, applying such a relationship to an entire ERT survey area likely leads to inaccurate estimations in some areas (Boaga et al., 2013). Further, time-lapse ERT data inversion can generate resistive anomaly artifacts below the wetting area and influence soil moisture estimation (Carey et al., 2017).

3.4. Hydraulic properties of the subsurface

In a ~38-ha watershed, Abdu et al. (2008) applied EMI to estimate soil properties such as clay content and soil water holding capacity, for which traditional low-level soil surveys can provide only limited information about their spatial patterns. They suggested that repeated EMI surveys could help with the identification of hydrological "hot spots" by contrasting EMI maps created during wet and dry periods (Abdu et al., 2008). On a slope covered by clayey soils, Perrone et al. (2008) combined ERT surveys, borehole samplings, and laboratory geotechnical tests to calculate pore water pressure and assess the stability of the studied slope. Results indicated that, while it could not give direct information on pore pressures, ERT was excellent at reconstructing the subsurface stratigraphy where information from boreholes was not sufficient. Reliable subsurface stratigraphy with well-defined boundaries could be used to map water pressure distributions and assess slope stability (Perrone et al., 2008). Moreover, based on the close relationship between soil cracks and bulk soil electrical resistivity (Ackerson et al., 2014), ERT surveys were conducted in a Vertisol to estimate soil crack porosity from inverted electrical resistivity, which matched reasonably with direct measurements by filling cracks with cement to obtain the volume of soil cracks (Ackerson et al., 2017).

In a semi-arid karstic region, Soupios et al. (2007) used vertical electrical sounding (VES), which measures the vertical distribution of resistivity in the subsurface) to estimate hydraulic parameters of the shallow aquifer (i.e., the lower boundary of the CZ). Their results indicated that VES coupled with pumping test data could accurately estimate aquifer parameters, including hydraulic conductivity, transmissivity, aquifer thickness, and transverse resistance. As the traditional drilling method of obtaining aquifer properties is often expensive, the authors suggested geophysical measurements as a cost-effective alternative for the study of aquifer parameters (Soupios et al., 2007). Bradford et al. (2009) compared the ability of 100-MHz GPR and neutron porosity logs to determine the porosity distribution of the saturated zone in a gravel bar. The GPR-derived porosity estimates differed from the neutron log estimates by less than 0.05. Mount and Comas (2014) used GPR to map the porosity distribution of the Miami Limestone. They compared GPR-estimated porosity with values obtained by direct analytical measurements and indirect image analysis. GPR proved to be an effective method to image the spatial variability of

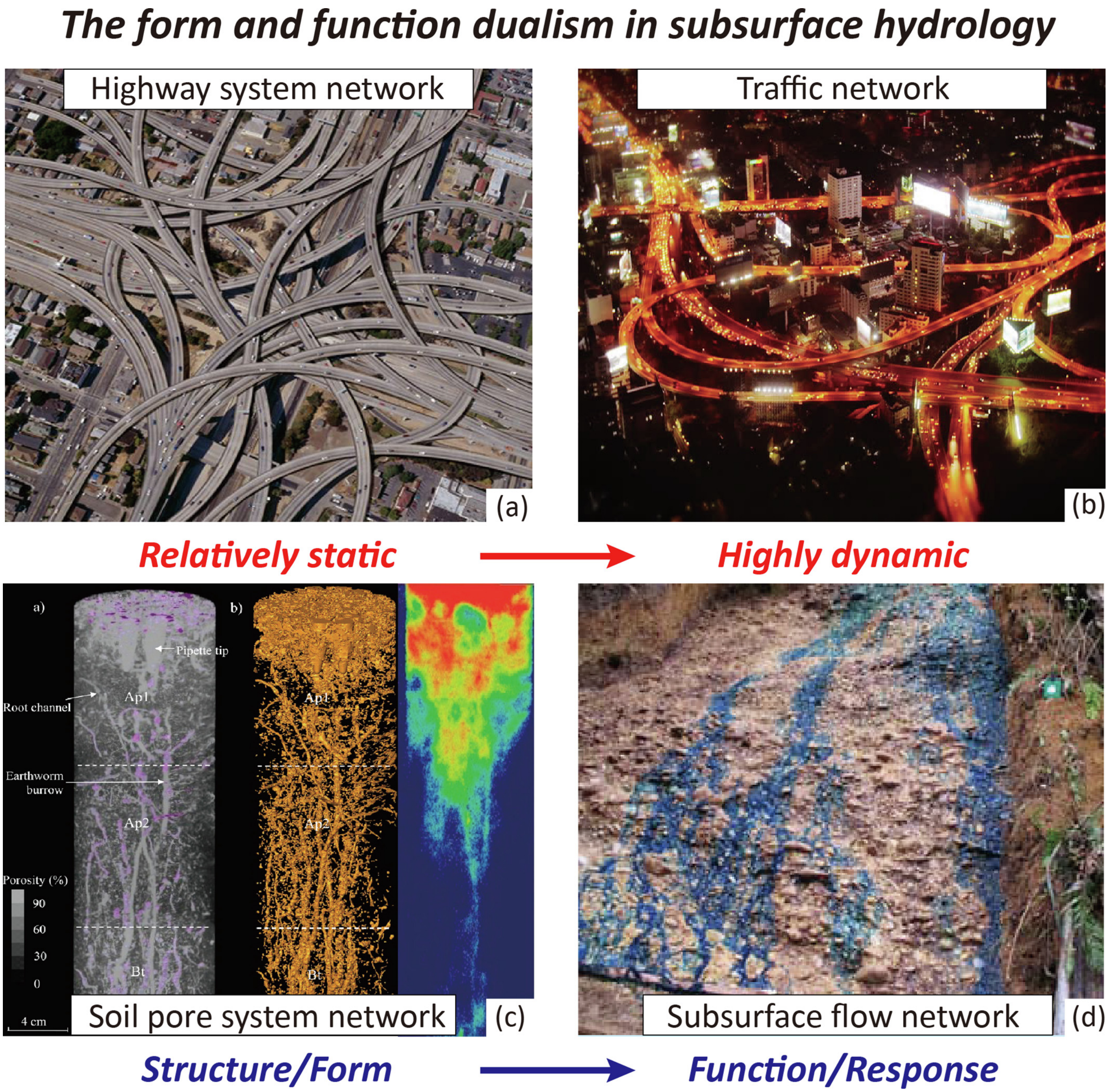

Fig. 6. The new conceptualization of non-uniform infiltration in porous soils. The macropore system is envisioned as a network of subsurface structures or forms, which are relatively static over time (a, c). After water infiltrates into the soils, a fraction of the macropore system is activated into a preferential flow network that represents the functions or responses of the subsurface, which is highly dynamic, just like the traffic on a highway (b, d). (c) Macropore networks (left and middle) and preferential flow pathways (right) revealed in a soil column by CT scanning (adapted, with permission, from Luo et al., 2010). (d) A lateral preferential flow network mapped by dye tracing on a hillslope (adapted, with permission, from Graham et al., 2010).

porosity in limestone and other karst aquifers (Mount and Comas, 2014).

4. A new conceptualization of non-uniform infiltration in structured soils

Various types of structures coexist in the subsurface, such as root channels, animal burrows, soil cracks, macropores, impermeable layers, interfaces between soil horizons, soil-bedrock interfaces, and fractures in weathered bedrock, all of which form interconnected networks of void spaces (Fig. 1c). As water infiltrates, often only a portion of the void space network is activated to transport water, which controls the dynamics of non-uniform flow (Guo et al., 2019; Luo et al., 2010). The interaction between void spaces and preferential flow can be envisioned as a "highway-traffic" relationship (Fig. 6c, d). The highway network (void spaces) is relatively static over time; whereas, the traffic on the highway (water flow) is highly dynamic, depending on the number of vehicles trying to use the system (water supply rate), the functions of intersections and traffic lights (the connectivity of the void spaces), road conditions (antecedent wetness of flow pathways), and the design of the highway system (the size and geometry of the void spaces; Guo and Lin, 2018; Fig. 6e, f). Therefore, we propose that the occurrence and dynamics of preferential flow following these preexisting structures and bypassing the bulk of the geomedia will dictate the non-uniform infiltration patterns in the real world. The essence of this conceptualization lies in the recognition of “form-and-function dualism” (Thompson, 1942), which describes the intimate link between subsurface structures and preferential infiltration dynamics. Establishing the relationship that links patterns, responses, and functions is imperative for describing and modeling a hydrological system (Sivapalan, 2005). Therefore, the combined observations of subsurface structures (form) and infiltration dynamics (function) by geophysical methods will help improve the prediction of infiltration. Ideally, such predictions will emphasize the continuum of macroscale behaviors and their links to the microscale nature of non-uniform flow processes (Assouline, 2013).

Characterizing form-and-function dualism in subsurface hydrological processes helps reveal the relationships between CZ structures and CZ services. Hydrology often triggers “hot spots” and “hot moments” of biogeochemical reactions and ecological functions. As a result, improving our understanding of the relationship between subsurface architectures and non-uniform infiltration behaviors would enhance our ability to determine chemical fluxes and elemental budgets in soils and ecosystems. Some research has examined form-and-function dualism at the soil core scale. For example, CT scanning results suggested that, under saturated conditions, overall macroporosity and the number of independent macropore pathways determine the magnitude of preferential flow. Meanwhile, saturated hydraulic conductivity, pathway number, hydraulic radius, and macropore angle determine its dispersivity (Luo et al., 2010; Katuwal et al., 2015; Fig. 6c). Under the unsaturated conditions, on the other hand, only a fraction of the macropore network may be activated with direct preferential flow. The geometric properties of this activated subset of macropores are distinctly different from the total macropore structure (Sammartino et al., 2012, 2015). Despite this small-scale success, establishing the form-and-function relationship between subsurface structures and infiltration dynamics remains challenging at the field scale. This is an area that has considerable potential to advance our understanding of real-world infiltration principles (Jackisch et al., 2017).

5. Future outlooks

Many previous studies determined infiltration with geophysical methods under laboratory conditions, such as in a sandbox with relatively uniform subsurface structures (e.g., Klenk et al., 2015; Mangel et al., 2015; Saintenoy et al., 2008). Due in part to their reduced resolution in field applications, uncertainty still exists in geophysical measurements of infiltration in the field, such as when inferring infiltration parameters or locating subsurface flow pathways (Beven and Germann, 2013; Binley et al., 2015). In addition, the performance of geophysical methods is site-dependent, so no standard field procedure has been developed to guide the geophysics-based observation or quantification of infiltration (Leucci, 2010). This makes cross-site comparisons of infiltration behaviors far more difficult and has delayed the formulation of infiltration principles applicable under various environments. In addition, the challenge of ground-truthing at large field scales has hampered the adoption of geophysical tools for these applications. Additional data (e.g., borehole information) are usually required to constrain the geophysical data, but such confirmation, especially at significant depths, is costly in terms of finances and labor, and is generally limited to a few point measurements. In some instances, invasive sampling is impossible or prohibited, in which conditions the interpretation of geophysical results is often unconvincing. All of these factors have impaired the resolution and precision of geophysics in the field investigation of infiltration. The rest of this section will be devoted to discussing some possible improvements or solutions to these problems.

5.1. *Integration of different geophysical methods*

One common problem with the use of geophysical methods is the ambiguity in the interpretation of their results. Rather than relying on expensive and spatially limited borehole data, multiple geophysical tools may be combined to generate more reliable interpretations (Al Hagrey, 2007). The advantage of combining multiple geophysical approaches is rooted in their different sensitivities and resolutions (Leucci and De Giorgi, 2005; Guo et al., 2020).

We propose different combinations of geophysical methods to enhance the field measurement of infiltration through the CZ. For example, GPR can be used to identify small subsurface structures (e.g., root channels and macropores), which can then be overlapped with ERT profiles to characterize soil layers and the depth to bedrock as well as the wetting areas, especially for deep zones where the signal-to-noise ratio is too low for effective GPR usage (e.g., Leucci, 2010; Guo et al., 2020). The conductivity values obtained from ERT can be used to estimate radar energy attenuation, allowing researchers to compensate for it and improve the quality of GPR data in these deep zones (Leucci and De Giorgi, 2005). Moreover, as GPR is relatively insensitive to vertical distributed structures, ERT surveys can be undertaken to constrain the underground geometry, especially the dipole-dipole array that is particularly good at mapping vertical patterns. Such ERT surveys also add information about groundwater table depth and clay distribution that may not be obtainable by GPR methods (Zenone et al., 2008). EMI surveys could then be conducted to track any GPR-identified structures, such as pipelines, across large-scale areas in urban and agricultural settings. At small scales, EMI data can be utilized to calibrate ERT measurements (van der Kruk et al., 2015) and to test multilayer inversion models (De Smedt et al., 2013a, b) that may be applied to locate flow-relevant structures at different depths. We advocate continuous research efforts, including the use of geophysical methods to measure subsurface structures and non-uniform infiltration patterns, to test and refine the form-and-function dualism of infiltration through the hidden half of the CZ. As summary of the promising points of application for combinations of GPR, ERT, and EMI to examine form-and-function dualism and to advance infiltration studies is compiled in Fig. 7.

In addition to near-surface (i.e., a few meters deep) geophysical tools (e.g., GPR, EMI, and ERT), geophysical techniques that can penetrate to a few tens of meters deep, such as seismic methods, have been adopted to image infiltration-related subsurface structures, e.g., fragipan horizon (Howard and Hickey, 2009), soil-saprolite interface and the interface between weathered and unweathered bedrock (Befus et al., 2011), and the fracturation of bedrock in the deeper CZ (Olona et al., 2010). Recent studies have signified the role of infiltration in the weathering of bedrock (Lebedeva and Brantley, 2020) as well as the importance of the water stored in fractured bedrock (also called rock moisture) in sustaining ecosystem during meteorological dry periods (Rempe and Dietrich, 2018). Geophysical techniques can contribute to a better understanding of the hydrological processes in the deeper layer of the CZ. There are also studies integrating seismic methods with near-surface geophysical techniques to enhance the measurement of infiltration. For example, Hilbich (2010) combined time-lapse refraction seismic tomography and ERT to detect permafrost degradation in the Swiss Alps, where the seismic method was more effective in identifying locations with ground ice degradation, and ERT, with a shorter sampling interval, could capture the relatively short-term dynamics in alpine permafrost. Another example is Holbrook et al. (2014), who combined seismic refraction and ERT to detect the variations in soil thickness and porosity across the catchment over granite bedrock. Their results suggested that saprolite is a crucial reservoir for water storage after rainfall infiltration. More field studies are required to test the integration of seismic methods with near-surface geophysics to improve the ability to determine infiltration in the deep subsurface.

5.2. *The new generation of geophysical equipment*

In general, a lower GPR frequency offers better depth penetration at

Assemblage of a virtual infiltrometer system

Flow-relevant structures		Non-uniform infiltration patterns	
Macropores:	3D GPR; GPR antenna array; 3D GPR+3D ERT	Flow path:	3D/2D GPR+3D/2D ERT; 2D GPR+2D ERT+2D EMI
Root system:	3D GPR; GPR antenna array; 3D GPR+3D ERT	Flow network:	3D GPR; GPR antenna array; 3D GPR+3D ERT+ 3D EMI
Soil layering:	3D/2D GPR+3D/2D ERT; 2D GPR+Multi-receiver EMI	2D moisture profile:	CMP GPR+2D GPR; 2D GPR+2D ERT
Impermeable layers:	3D/2D GPR+3D/2D ERT; 2D GPR+Multi-receiver EMI	3D moisture profile:	3D GPR+3D ERT
Regolith thickness:	3D/2D GPR+3D/2D ERT	Surface moisture:	3D GPR+3D EMI; Ground-wave GPR+EMI; GPR antenna array+EMI
Bedding fabric:	3D/2D GPR+3D/2D ERT	Wetting front:	3D/2D GPR+3D/2D ERT; GPR antenna array
Fractures & joints:	3D/2D GPR+3D/2D ERT	Perched water table:	3D/2D GPR+3D/2D ERT
Drainage system:	3D GPR+3D ERT; 2D GPR+2D EMI		

Fig. 7. Possible combinations of different geophysical methods as a virtual infiltrometer system to validate and refine the form-and-function dualism approach for infiltration monitoring. These combinations can enhance the detection of flow-relevant structures (left) and help researchers visualize non-uniform infiltration patterns in structured geomedia (right).

the cost of worse spatial resolution (Parsekian et al., 2015). One proven method to negate this tradeoff between detection resolution and penetration depth involves conducting GPR surveys using antennas with a range of frequencies (De Coster and Lambot, 2016). The integration of GPR data collected at multiple frequencies allows the combination of high-resolution data with extensive penetration depth (Cui et al., 2020). This data collection approach also helps separate the actual signals of targets from radar image artifacts, since random artifacts rarely recur on multi-frequency images. Recently, the development of the dual-frequency GPR systems that simultaneously collect data at two frequencies opens up new opportunities to obtain high-resolution data for both shallow and deep targets by saving time and ensuring that the scans are contemporaneous (Klenk et al., 2015). In addition to the direct comparison between multi-frequency images, there are also studies working on the multi-frequency data fusion algorithm to maximize both resolution and characterization depth for retrieving useful information (e.g., De Coster and Lambot, 2018; Xu et al., 2019). Moreover, the portable rotary laser positioning system (RLPS) has been integrated with GPR into a 3D imaging system that permits the real-time recording of coordinates with centimeter-scale accuracy (e.g., Truss et al., 2007). The recent improvements to 3D radar arrays (e.g., 3D Mala GPR Imaging Radar Array), which consist of several transmitting and receiving antennas, has boosted the rapid collection of full-resolution 3D data, which can be analyzed to monitor infiltration in the field.

Apart from new GPR systems, Chambers et al. (2016) has developed and refined a long-term, low-cost, and real-time ERT system that can be deployed for months at a time to monitor 3D electrical resistivity dynamics associated with the full cycles of wetting and drying over many infiltration events. This system includes a remote access data acquisition software that allows convenience to set the start time of a survey and the frequency of surveys in any day and provides an output file compatible with the resistivity reconstruction algorithm. Besides, a wireless data transmission system is included as well, which uses the 3G network that permits real-time data review and download. Regarding EMI technique, the development of multi-receiver EMI sensors and the multi-layer inversion modeling algorithms has allowed EMI to generate information at different depths (De Smedt et al., 2013a, b; Saey et al., 2012), which enhances the application of EMI to study infiltration, e.g., locating infiltration-relevant structures and generating soil moisture maps for different depths.

5.3. Extension from static surveys to dynamic investigations

In most current applications, geophysical data are used as a proxy for the spatial distribution of hydrologically relevant properties, with the result that the flow of information is in one direction: from geophysics to hydrology (Binley et al., 2015). However, as recently reported in Jackisch et al. (2017) and Angermann et al. (2017), in subsurface environments with complex structures, the overall heterogeneity would likely obscure the identification of flow-relevant structures and impair the inversion of hydrologic properties under static conditions, especially without a shift from the non-active (no infiltration condition) to active hydrologic states (infiltration occurring condition). To address this issue, a coupled geophysical inversion with feedback between geophysics and hydrology may be considered (Binley et al., 2015). Dynamics investigations under hydrologically active conditions (e.g., after rainfall or irrigation) are suggested to help highlight the flow-relevant structures (Di Prima et al., 2020). For example, Zhang et al. (2014) found that if only a static geophysical image was used, it was difficult to distinguish GPR reflections produced by the interfaces between soil layers from those caused by local water accumulation; whereas, based on the difference in GPR reflections between the dry and wet seasons, the identification of a water-restrictive in the deep soils and the soil-saprock interface in the shallow soils became more convincing. This reflects a flow of information from hydrology to geophysics. Current research into mapping soil layering patterns with GPR has also demonstrated the effectiveness of this coupled interpretation of geophysical results (Romero-Ruiz et al., 2018). This combined assessment of form and function can reduce the ambiguity of the interpretation and inversion of geophysical data (Binley et al., 2015; Jackisch et al., 2017).

5.4. Enhancing the processing of time-lapse GPR data

Signal differences between time-lapse GPR images are the key to locating subsurface flowpaths. However, variability in background noise and signal drift from changing conditions can cause signal differences, which affects the detection of flowpaths. Therefore, researchers need to refine their data post-processing techniques to improve the reproducibility of time-lapse GPR images and to develop new indicators to detect moisture changes in GPR images. Nyquist et al. (2014) proposed the dynamic time warping (DTW) method to compensate for the time shift at each surface sampling location. For example, if $X(x_1, x_2, \ldots, x_n)$ and $Y(y_1, y_2, \ldots, y_n)$ are two time-lapse traces of the same length, n, then an $n \times n$ matrix, M, can be defined to represent the point-to-point correspondence between X and Y, where $M_{i,j}$ indicates the distance, $d(x_i, y_j)$, between x_i and y_j. The matching relationship between X and Y can be represented by a time warping path, $W = (w_1, w_2, \ldots, w_k)$, where $w_k = (i, j)$ indicates the point-to-point relationship between x_i and y_j. If a path is the lowest cost path between two traces, the corresponding DTW distance can be quantified as

$$DTW(X, Y) = \min\left(\frac{1}{K}\sum_{k=1}^{K} d_k,\ W = (w_1, w_2, \ldots, w_k)\right) \tag{1}$$

where $d_k = d(x_i, y_j)$ indicates the distance represented as $w_k = (i, j)$ on W that can be used to align Y to X.

Geometric correction methods that are typically used to process satellite images can also be applied to standardize GPR images, such as the least square transformation approach to align ground control points (GCPs; Toutin, 2011). Possible GCPs in GPR images include hyperbolic reflections generated by point reflectors, either natural or emplaced, or horizontal reflections by subsurface stratification. GPR image distortion can be modeled by a mapping the transformation from subsequent image coordinates (i.e., surface distance and depth) to background image coordinates of selected GCPs. Denoting subsequent image coordinates as $r = (x, y)$ and background coordinates as $r' = (x', y')$, transformation functions are given by the equations

$$x' = f_1(x, y) = \sum_{j=0}^{q}\sum_{k=0}^{q-1} a_{j,k}x^j y^k \text{ and} \tag{2}$$

$$y' = f_2(x, y) = \sum_{j=0}^{q}\sum_{k=0}^{q-1} b_{j,k}x^j y^k \tag{3}$$

where q is the order of the polynomial equation, and $a_{j,\,k}$ and $b_{j,\,k}$ are coefficients derived from the least square function to fit the coordinates of a set of GCPs between two images. The variables i and j indicate the ith and jth GCP, respectively. Using f_1 and f_2, the subsequent image can be corrected to the same geometry as the background image. This improved reproducibility suppresses geometric mismatch between time-lapse images and ensures proper correspondence between signal changes in the time-lapse images and the areas with actual moisture changes.

Subsurface flow increases the reflection amplitude and causes a time-shift in the GPR signal below the wetting zone, due to the reduced wave velocity (Truss et al., 2007). Both of these phenomena can be used to identify signal changes. However, field GPR data is often complex, and reflection amplitude is influenced by antenna-ground coupling, so GPR studies have been unable to effectively capitalize on amplitude data. Motivated by advanced image-processing methods, Allroggen and Tronicke (2016) developed two attributes to distinguish amplitude similarity (*AS*) and structural similarity (*SS*; related to time-shift) variations in time-lapse GPR images:

$$AS(x, y) = \frac{2\sigma_x\sigma_y + a_1}{\sigma_x^2 + \sigma_y^2 + a_1},\ SS(x, y) = \frac{\sigma_{xy} + a_2}{\sigma_x\sigma_y + a_2} \tag{4}$$

where σ indicates the standard deviation of the amplitude sequence of a GPR trace collected at two acquisition times (x, y), σ_{xy} is the covariance of the amplitude sequences at two time-steps, and a_1 and a_2 are constants that prevent near-zero σ values, which causes numerical instability. The GPR attributes are calculated with 1 being highly similar and -1 being the most dissimilar. Recent studies have demonstrated the efficacy of *SS* to identify areas with moisture changes (Angermann et al., 2017; Guo et al., 2020). However, the ability of *AS* to indicate GPR signal changes remains unclear. Future studies should improve the capacity of *AS* to quantify GPR signal differences. For example, *AS* and *SS* could be combined into a new comprehensive similarity indicator (*CS*) to measure overall signal changes between time-lapse GPR images:

$$CS(x, y) = [AS(x, y)]^{\alpha}\cdot[SS(x, y)]^{\beta} \tag{5}$$

where $\alpha > 0$ and $\beta > 0$ are two weighting factors, and (x, y) indicates two acquisition times. By setting both α and β as 1 in Eq. (5), Allroggen et al. (2020) calculated the values of *CS* for time-lapse GPR data collected after an infiltration experiment on a periglacial slope. Their results confirmed the efficacy of *CS* to monitor subsurface flow processes and the dynamics of the water table. Continued efforts are required to refine the calculation of *CS* by testing different combinations of the weighting factors (α and β) to further enhance the efficiency of time-lapse GPR to study infiltration processes.

6. Summary and conclusion

This review was motivated by a long-standing need in the hydrologic sciences to enhance the field determination of non-uniform infiltration through structured geomedia. This ability would aid in the improvement of infiltration theory that addresses the intimate link between flow dynamics and structural heterogeneity in the subsurface of the CZ. Infiltration is a fundamentally important hydrologic process that influences a variety of soil, water, and ecosystem processes in the CZ. However, our ability to reveal the complex and highly dynamic process of infiltration, especially in hidden structured subsurface (soils and regoliths), continues to languish. Classical infiltration theories were built on assumptions of uniform flow, and steady-state measurements at the ground surface ignore the realities of structured subsurface media and preferential flow within the CZ. A more robust and realistic infiltration principle, which takes into account the transient and heterogeneous nature of subsurface flow in diverse soils and landscapes, is needed. A suite of geophysical techniques, especially GPR, ERT, and EMI, have opened up new possibilities for minimally invasive determination of infiltration patterns across time and space. We have reviewed the applications of geophysical techniques in imaging subsurface infiltration-relevant structures, monitoring non-uniform infiltration processes in structured geomedia, and mapping the spatial distribution of soil moisture and subsurface hydrologic properties. Moreover, we have proposed a new and realistic principle of subsurface hydrology that addresses non-uniform or preferential infiltration in structured geomedia: form-and-function dualism. Continuous improvement in the innovative visualization and quantification of non-uniform or preferential infiltration will advance hydrologic sciences, as well as the related fields of soil and CZ sciences.

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References

Abdu, H., Robinson, D.A., Seyfried, M., Jones, S.B., 2008. Geophysical imaging of watershed subsurface patterns and prediction of soil texture and water holding capacity. Water Resour. Res. 44 (4), 1–10.

Ackerson, J., McInnes, K., Morgan, C., Everett, M., 2017. Measuring crack porosity using three-dimensional electrical resistivity tomography. Soil Sci. Soc. Am. J. 81, 1025–1035.

Ackerson, J., Morgan, C., Everett, M., McInnes, K., 2014. The role of water content in electrical resistivity tomography of a Vertisol. Soil Sci. Soc. Am. J. 78, 1552–1562.
Al Hagrey, S.A., 2007. Geophysical imaging of root-zone, trunk, and moisture heterogeneity. J. Exp. Bot. 58 (4), 839–854.
Allred, B., Wishart, D., Martinez, L., Schomberg, H., Mirsky, S., Meyers, G., Elliott, J., Charyton, C., 2018. Delineation of agricultural drainage pipe patterns using ground penetrating radar integrated with a real-time kinematic global navigation satellite system. Agriculture 8 (11), 1–14.
Allroggen, N., Tronicke, J., 2016. Attribute-based analysis of time-lapse ground-penetrating radar data. Geophysics 81 (1), H1–H8.
Allroggen, N., van Schaik, N.L.M.B., Tronicke, J., 2015. 4D ground-penetrating radar during a plot scale dye tracer experiment. J. Appl. Geophys. 118 (2015), 139–144.
Allroggen, N., Beiter, D., Tronicke, J., 2020. Ground-penetrating radar monitoring of fast subsurface processes. Geophysics 85 (3), 1MJ-Z13. https://doi.org/10.1190/geo2019-0737.1.
Angermann, L., Jackisch, C., Allroggen, N., Sprenger, M., Zehe, E., Tronicke, J., Weiler, M., Blume, T., 2017. Form and function in hillslope hydrology: characterization of subsurface flow based on response observations. Hydrol. Earth Syst. Sci. 21, 3727–3748.
Armstrong, R.T., McClure, J.E., Berrill, M.A., Rücker, M., Schlüter, S., Berg, S., 2016. Beyond Darcy's law: The role of phase topology and ganglion dynamics for two-fluid flow. Phys. Rev. E 94 (4–1).
Assouline, S., 2013. Infiltration into soils: Conceptual approaches and solutions. Water Resour. Res. 49 (4), 1755–1772.
Befus, K.M., Sheehan, A.F., Leopold, M., Anderson, S.P., Anderson, R.S., 2011. Seismic constraints on Critical Zone architecture, Boulder Creek Watershed, Front Range, Colorado. Vadose Zone J. 10, 915–927.
Beven, K., Germann, P., 2013. Macropores and water flow in soils revisited. Water Resour. Res. 49, 3071–3092.
Binley, A., Kemna, A., 2005. DC resistivity and induced polarization methods. In: Rubin, Y., Hubbard, S.S. (Eds.), Hydrogeophysics. Springer Netherlands, Dordrecht.
Binley, A., Hubbard, S.S., Huisman, J.A., Revil, A., Robinson, D.A., Singha, K., Slater, L.D., 2015. The emergence of hydrogeophysics for improved understanding of subsurface processes over multiple scales. Water Resour. Res. 51, 3837–3866.
Boaga, J., Rossi, M., Cassiani, G., 2013. Monitoring soil-plant interactions in an apple orchard using 3D electrical resistivity tomography. Procedia Environ. Sci. 19, 394–402.
Borden, K.A., Thomas, S.C., Isaac, M.E., 2017. Interspecific variation of tree root architecture in a temperate agroforestry system characterized using ground-penetrating radar. Plant Soil 410 (1–2), 323–334.
Bovi, R., Moreira, C., Rosolen, V., Rosa, F., Furlan, L., Helene, L., 2020. Piping process: Genesis and network characterization through a pedological and geophysical approach. Geoderma 361, 1–14.
Bradford, J.H., Clement, W.P., Barrash, W., 2009. Estimating porosity with ground-penetrating radar reflection tomography: a controlled 3-D experiment at the Boise Hydrogeophysical Research Site. Water Resour. Res. 45, 1–11.
Bradford, J.M., Ferris, J.E., Remley, P.A., 1987. Interrill soil erosion processes: I. effect of surface sealing on infiltration, runoff, and soil splash detachment. Soil Sci. Soc. Am. J. 51, 1566–1571.
Butnor, J.R., Doolittle, J.A., Kress, L., Cohen, S., Johnsen, K.H., 2001. Use of ground-penetrating radar to study tree roots in the southeastern United States. Tree Physiol. 21 (17), 1269–1278.
Butnor, J.R., Samuelson, L., Stokes, T.A., Johnsen, K.H., Anderson, P.H., González-Benecke, C.A., 2016. Surface-based GPR underestimates below-stump root biomass. Plant Soil 402, 47–62.
Carey, A., Paige, G., Carr, B., Dogan, M., 2017. Forward modeling to investigate inversion artifacts resulting from time-lapse electrical resistivity tomography during rainfall simulations. J. Appl. Geophys. 145, 39–49.
Chambers, J.E., Meldrum, P.I., Wilkinson, P.B., Gunn, D., Uhlemann, S., Kuras, O., Swift, R., Inauen, C., Butler, S., 2016. Remote condition assessment of geotechnical assets using a new low-power ERT monitoring system. In: 22nd European Meeting of Environmental and Engineering Geophysics, Barcelona, Spain, September 2016.
Chambers, J.E., Gunn, D.A., Wilkinson, P.B., Meldrum, P.I., Haslam, E., Holyoake, S., Kirkham, M., Kuras, O., Merritt, A., Wragg, J., 2014. 4D electrical resistivity tomography monitoring of soil moisture dynamics in an operational railway embankment. Near Surf. Geophys. 12, 61–72.
Comas, X., Kettridge, N., Binley, A., Slater, L., Parsekian, A., Baird, A.J., Strack, M., Waddington, J.M., 2014. The effect of peat structure on the spatial distribution of biogenic gases within bogs. Hydrol. Process. 28 (22), 5483–5494.
Corwin, D.L., Lesch, S.M., 2005. Apparent soil electrical conductivity measurements in agriculture. Comput. Electron. Agric. 46 (1–3), 11–43.
Cui, X., Liu, X., Cao, X., Fan, B., Zhang, Z., Chen, J., Chen, X., Guo, L., 2020. Pairing dual-frequency GPR in summer and winter enhances the detection and mapping of coarse roots in the semi-arid shrubland in China. Eur. J. Soil Sci. 71, 236–251.
Damiano, E., Olivares, L., 2010. The role of infiltration processes in steep slope stability of pyroclastic granular soils: laboratory and numerical investigation. Nat. Hazards 52, 329–350.
De Coster, A., Lambot, S., 2016. Multi-frequency GPR data fusion. In: 2016 16th International Conference on Ground Penetrating Radar (GPR), IEEE, Hong Kong, China, June 13–16 2016, pp. 1–6.
De Coster, A., Lambot, S., 2018. Fusion of multifrequency GPR data freed from antenna effects. IEEE J. Selected Topics Appl. Earth Observations Remote Sensing, 11, 664–674.
De Smedt, P., Van Meirvenne, M., Davies, N.S., Bats, M., Saey, T., De Reu, J., Meerschman, E., Gelorini, V., Zwertvaegher, A., Antrop, M., et al., 2013a. A multi-disciplinary approach to reconstructing Late Glacial and Early Holocene landscapes. J. Archaeol. Sci., 40(2), 1260–1267.
De Carlo, L., Battilani, A., Solimando, D., Caputo, M., 2020. Application of time-lapse ERT to determine the impact of using brackish wastewater for maize irrigation. J. Hydrol. 582, 1–9.
De Smedt, P., Saey, T., Lehouck, A., Stichelbaut, B., Meerschman, E., Islam, M.M., Van De Vijver, E., Van Meirvenne, M., 2013b. Exploring the potential of multi-receiver EMI survey for geoarchaeological prospection: a 90 ha dataset. Geoderma 199, 30–36.
Deiana, R., Cassiani, G., Villa, A., Bagliani, A., Bruno, V., 2008. Calibration of a vadose zone model using water injection monitored by GPR and electrical resistance tomography. Vadose Zone J. 7 (1), 215–226.
Di Prima, S., Winiarski, T., Angulo-Jaramillo, R., Stewart, R.D., Castellini, M., Abou Najm, M.R., Ventrella, D., Pirastru, M., Giadrossich, F., Capello, G., Biddoccu, M., Lassabatere, L., 2020. Detecting infiltrated water and preferential flow pathways through time-lapse ground-penetrating radar surveys. Sci. Total Environ., 726: 1–16. doi: 10.1016/j.scitotenv.2020.138511.
Dietrich, S., Weinzettel, P.A., Varni, M., 2014. Infiltration and drainage analysis in a heterogeneous soil by electrical resistivity tomography. Soil Sci. Soc. Am. J. 78 (4), 1153–1167.
Donohue, S., McCarthy, V., Rafferty, P., Orr, A., Flynn, R., 2015. Geophysical and hydrogeological characterization of the impacts of on-site wastewater treatment discharge to groundwater in a poorly productive bedrock aquifer. Sci. Total Environ. 523, 109–119.
Doolittle, J., Zhu, Q., Zhang, J., Guo, L., Lin, H., 2012. Geophysical investigations of soil-landscape architecture and its impacts on subsurface flow. In: Lin, H. (Eds.) Hydropedology: synergistic integration of soil science and hydrology. Elsevier B V, Amsterdam, Netherlands, pp. 413–447.
Dou, Q.X., Wei, L.J., Magee, D.R., Cohn, A.G., 2017. Real-time hyperbola recognition and fitting in GPR data. IEEE Trans. Geosci. Remote Sens. 55, 51–62.
Eldridge, D., Zaady, E., Shachak, M., 2000. Infiltration through three contrasting biological soil crusts in patterned landscapes in the Negev, Israel. Catena 40 (3), 323–336.
Fan, Y., Miguez-Macho, G., Jobbagy, E.G., Jackson, R.B., Otero-Casal, C., 2017. Hydrologic regulation of plant rooting depth. Proc. National Acad. Sci. USA, 114, 10572–10577.
Gormally, K.H., McIntosh, M.S., Mucciardi, A.N., McCarty, G.W., 2011. Ground-penetrating radar detection and three-dimensional mapping of lateral macropores: II. riparian application. Soil Sci. Soc. Am. J. 75 (4), 1236–1243.
Graham, C.B., Woods, R.A., McDonnell, J.J., 2010. Hillslope threshold response to rainfall: (1) a field based forensic approach. J. Hydrol. 393 (2010), 65–76.
Green, W.H., Ampt, G.A., 1911. Studies on soil physics Part I - the flow of air and water through soils. J. Agric. Sci. 4 (1), 1–24.
Greggio, N., Giambastiani, B., Balugani, E., Amaini, C., Antonellini, M, 2018. High-resolution electrical resistivity tomography (ERT) to characterize the spatial extension of freshwater lenses in a salinized coastal aquifer. Water 10, 1–16.
Guo, L., Chen, J., Lin, H., 2014. Subsurface lateral preferential flow network revealed by time-lapse ground-penetrating radar in a hillslope. Water Resour. Res. 50, 9127–9147.
Guo, L., Lin, H., 2016. Critical zone research and observatories: current status and future perspectives. Vadose Zone J. 15 (9), 1–13.
Guo, L., Lin, H., 2018. Addressing two bottlenecks to advance the understanding of preferential flow in soils. Adv. Agron. 147, 61–117.
Guo, L., Wu, Y., Chen, J., Hirano, Y., Tanikawa, T., Li, W., Cui, X., 2015. Calibrating the impact of root orientation on root quantification using ground-penetrating radar. Plant Soil 395 (1–2), 289–305.
Guo, L., Lin, H., Fan, B., Nyquist, J., Toran, L., Mount, G., 2019. Preferential flow through shallow fractured bedrock and a 3D fill-and-spill model of hillslope subsurface hydrology. J. Hydrol. 576 (2019), 430–442.
Guo, L., Mount, G., Hudson, S., Lin, H., Levia, D., 2020. Pairing geophysical techniques improves understanding of the near-surface critical zone: visualization of preferential routing of stemflow along coarse roots. Geoderma 357, 1–12. https://doi.org/10.1016/j.geoderma.2019.113953.
Haarder, E.B., Looms, M.C., Jensen, K.H., Nielsen, L., 2011. Visualizing unsaturated flow phenomena using high-resolution reflection ground penetrating radar. Vadose Zone J. 10 (1), 84–97.
Han, X., Liu, J., Zhang, J., Zhang, Z., 2016. Identifying soil structure along headwater hillslopes using ground penetrating radar based technique. J. Mountain Sci. 13, 405–415.
Hilbich, C., 2010. Time-lapse refraction seismic tomography for the detection of ground ice degradation. The Cryosphere 4, 243–259.
Hillel, D., 1998. Environmental soil physics. Academic Press, New York.
Hirano, Y., Yamamoto, R., Dannoura, M., Kenji, Aono, Igarashi, T., Ishii, M., Yamase, K., Makita, N., Kanazawa, Y., 2012. Detection frequency of Pinus thunbergii roots by ground-penetrating radar is related to root biomass. Plant Soil 360, 363–373.
Holbrook, W.S., Riebe, C.S., Elwaseif, M., Hayes, J.L., Basler-Reeder, K., Harry, D.L., Malazian, A., Dosseto, A., Hartsough, P.C., Hopmans, J.W., 2014. Geophysical constraints on deep weathering and water storage potential in the Southern Sierra Critical Zone Observatory. Earth Surf. Proc. Land. 39, 366–380.
Holden, J., 2004. Hydrological connectivity of soil pipes determined by ground-penetrating radar tracer detection. Earth Surf. Proc. Land. 29, 437–442.
Horton, R.E., 1941. An approach toward a physical interpretation of infiltration-capacity. Soil Sci. Soc. Am. J. 5 (C), 399.
Howard, W., Hickey, C.J., 2009. Investigation of the near subsurface using acoustic to seismic coupling. Ecohydrology 2, 263–269.
Huang, M., Lee Barbour, S., Elshorbagy, A., Zettl, J.D., Cheng Si, B., 2011. Infiltration and drainage processes in multi-layered coarse soils. Can. J. Soil Sci. 91 (2), 169–183.
Huisman, J.A., Hubbard, S.S., Redman, J.D., Annan, A.P., 2003. Measuring soil water content with ground penetrating radar: a review. Vadose Zone J. 2, 476–491.

Illawathure, C., Parkin, G., Lambot, S., Galagedara, L., 2020. Evaluating soil moisture estimation from ground-penetrating radar hyperbola fitting with respect to a systematic time-domain reflectometry data collection in a boreal podzolic agricultural field. Hydrol. Process. 2020, 1–18. https://doi.org/10.1002/hyp.13646.
Jackisch, C., Angermann, L., Allroggen, N., Sprenger, M., Blume, T., Tronicke, J., Zehe, E., 2017. Form and function in hillslope hydrology: in situ imaging and characterization of flow-relevant structures. Hydrol. Earth Syst. Sci. 21, 3749–3775.
Jayawickreme, D.H., Santoni, C.S., Kim, J.H., Jobbágy, E.G., Jackson, R.B., 2011. Changes in hydrology and salinity accompanying a century of agricultural conversion in Argentina. Ecol. Appl. 21 (7), 2367–2379.
Katuwal, S., Moldrup, P., Lamandé, M., Tuller, M., de Jonge, L.W., 2015. Effects of CT number derived matrix density on preferential flow and transport in a macroporous agricultural soil. Vadose Zone J. 14 (7), 1–13.
Kinlaw, A., Grasmueck, M., 2012. Evidence for and geomorphologic consequences of a reptilian ecosystem engineer: The burrowing cascade initiated by the Gopher Tortoise. Geomorphology 157–158, 108–121.
Klenk, P., Jaumann, S., Roth, K., 2015. Monitoring infiltration processes with high-resolution surface-based Ground-Penetrating Radar. Hydrol. Earth Syst. Sci. Discuss. 12, 12215–12246.
Klotzsche, A., Jonard, F., Looms, M.C., van der Kruk, J., Huisman, J.A., 2018. Measuring soil water content with ground penetrating radar: a decade of progress. Vadose Zone J. 17 (1), 1–9.
Koyama, C., Liu, H., Takahashi, K., Shimada, M., Watanabe, M., Khuut, T., Sato, M., 2017. In-Situ measurement of soil permittivity at various depths for the calibration and validation of low-frequency SAR soil moisture models by using GPR. Remote Sensing 9, 1–14. https://doi.org/10.3390/rs9060580.
Lebedeva, M.I., Brantley, S.L., 2020. Relating the depth of the water table to depth of weathering. Earth Surf. Processes Landforms, in press. https://doi.org/10.1002/esp.4873.
Leslie, I.N., Heinse, R., 2013. Characterizing soil-pipe networks with pseudo-three-dimensional resistivity tomography on forested hillslopes with restrictive horizons. Vadose Zone J. 12 (4), 1–10.
Leucci, G., 2010. The use of three geophysical methods for 3D images of total root volume of soil in urban environments. Explor. Geophys. 41 (4), 268–278.
Leucci, G., De Giorgi, L., 2005. Integrated geophysical surveys to assess the structural conditions of a karstic cave of archaeological importance. Nat. Hazard. Earth Syst. Sci. 5 (1), 17–22.
Li, X., Yang, Z., Li, Y., Lin, H., 2009. Connecting ecohydrology and hydropedology in desert shrubs: stemflow as a source of preferential flow in soils. Hydrology Earth Syst. Sci. 13 (7), 1133–1144.
Lin, H., 2010. Linking principles of soil formation and flow regimes. J. Hydrol. 393 (1–2), 3–19.
Liu, X., Cui, X., Guo, L., Chen, J., Li, W., Yang, D., Cao, X., Chen, X., Liu, Q., 2019. Estimating soil water content from coarse root reflections on ground-penetrating radar radargrams. Plant Soil 436, 623–639.
Liu, X., Chen, J., Butnor, J., Qin, G., Cui, X., Fan, B., Lin, H., Guo, L., 2020. Non-invasive 2D and 3D mapping of root zone soil moisture through the detection of coarse roots with ground-penetrating radar. Water Resources Res., in press. https://doi.org/10.1029/2019WR026930.
Luo, L., Lin, H., Schmidt, J., 2010. Quantitative relationships between soil macropore characteristics and preferential flow and transport. Soil Sci. Soc. Am. 74 (6), 1929–1937.
Ma, Y., Zhang, Y., Zubrzycki, S., Guo, Y., Bin, Farhan S., 2015. Hillslope-scale variability in seasonal frost depth and soil water content investigated by GPR on the southern margin of the sporadic permafrost zone on the Tibetan Plateau. Permafrost Periglac. Process. 26 (4), 321–334.
Mangel, A.R., Lytle, B.A., Moysey, S.M.J., 2015. Automated high-resolution GPR data collection for monitoring dynamic hydrologic processes in two and three dimensions. Lead. Edge 34 (2), 190–196.
Mangel, A.R., Moysey, S.M.J., Bradford, J., 2020. Reflection tomography of time-lapse GPR data for studying dynamic unsaturated flow phenomena. Hydrol. Earth Syst. Sci. 24, 159–167.
Michot, D., Benderitter, Y., Dorigny, A., Nicoullaud, B., King, D., Tabbagh, A., 2003. Spatial and temporal monitoring of soil water content with an irrigated corn crop cover using surface electrical resistivity tomography. Water Resour. Res. 39 (5), 1–14.
Molon, M., Boyce, J.I., Arain, M.A., 2017. Quantitative, nondestructive estimates of coarse root biomass in a temperate pine forest using 3-D ground-penetrating radar (GPR). J. Geophys. Res.-Biogeosci. 122, 80–102.
Mount, G.J., Comas, X., 2014. Estimating porosity and solid dielectric permittivity in the Miami Limestone using high-frequency ground penetrating radar (GPR) measurements at the laboratory scale. Water Resour. Res. 50 (10), 7590–7605.
Nichols, P., McCallum, A., Lucke, T., 2017. Using ground penetrating radar to locate and categorise tree roots under urban pavements. Urban For. Urban Greening 27, 9–14.
Nyquist, J.E., Toran, L., Pitman, L., Lin, H., 2014. Dynamic time warping of time-lapse GPR data to monitor infiltration at the Shale Hills Critical Zone Observatory. In: Symposium for the Application of Geophysics to Environmental and Engineering Problems, Boston, MA.
Nyquist, J.E., Toran, L., Pitman, L., Guo, L., Lin, H., 2018. Testing the fill-and-spill model of subsurface lateral flow using ground-penetrating radar and dye tracing. Vadose Zone J. 17 (1), 1–13.
Ogden, F.L., Lai, W., Steinke, R.C., Zhu, J., Talbot, C.A., Wilson, J.L., 2015. A new general 1-D vadose zone flow solution method. Water Resour. Res. 51 (6), 4282–4300.
Ohashi, M., Ikeno, H., Sekihara, K., Tanikawa, T., Dannoura, M., Yamase, K., Todo, C., Tomita, T., Hirano, Y., 2019. Reconstruction of root systems in Cryptomeria japonica using root point coordinates and diameters. Planta 249 (2), 445–455.
Olona, J., Pulgar, J.A., Fernandez-Viejo, G., Lopez-Fernandez, C., Gonzalez-Cortina, J.M., 2010. Weathering variations in a granitic massif and related geotechnical properties through seismic and electrical resistivity methods. Near Surf. Geophys. 8, 585–599.
Parlange, J.Y., 1971. Theory of water-movement in soils: 2. One-dimensional infiltration. Soil Sci. 111 (3), 170–174.
Parsekian, A.D., Slater, L., Comas, X., Glaser, P.H., 2010. Variations in free-phase gases in peat landforms determined by ground-penetrating radar. J. Geophys. Res. Biogeosci. 115 (G2), 1–13.
Parsekian, A.D., Singha, K., Minsley, B.J., Holbrook, W.S., Slater, L., 2015. Multiscale geophysical imaging of the critical zone. Rev. Geophys. 53 (1), 1–26.
Perrone, A., Vassallo, R., Lapenna, V., Maio, C. Di, 2008. Pore water pressures and slope stability: a joint geophysical and geotechnical analysis. J. Geophys. Eng. 5(3), 323–337.
Philip, J.R., 1969. Theory of infiltration. Adv. Hydrosci. 85 (5), 215–296.
Rawls, W.J., Brakensiek, D.L., Soni, B., 1983. Agricultural management effects on soil water processes Part I: soil water retention and Green and Ampt infiltration parameters. Trans. Am. Soc. Agric. Eng. 26 (6), 1747–1752.
Rempe, D.M., Dietrich, W.E., 2018. Direct observations of rock moisture, a hidden component of the hydrologic cycle. Proc. Natl. Acad. Sci. 115 (11), 2664–2669.
Rezanezhad, F., Vogel, H.-J., Roth, K., 2006. Experimental study of fingered flow through initially dry sand. Hydrol. Earth Syst. Sci. Discuss. 3 (4), 2595–2620.
Robinson, D.A., Abdu, H., Lebron, I., Jones, S.B., 2012. Imaging of hill-slope soil moisture wetting patterns in a semi-arid oak savanna catchment using time-lapse electromagnetic induction. J. Hydrol. 416–417, 39–49.
Robinson, D.A., Binley, A., Crook, N., Day-Lewis, F.D., Ferre, T.P.A., Grauch, V.J.S., Knight, R., Knoll, M., Lakshmi, V., Miller, R., Nyquist, J., Pellerin, L., Singha, K., Slater, L., 2008. Advancing process-based watershed hydrological research using near-surface geophysics: a vision for, and review of, electrical and magnetic geophysical methods. Hydrol. Processes, 22(18), 3604–3635.
Robinson, J., Buda, A., Collick, A., Shober, A., Ntarlagiannis, D., Bryant, R., Folmar, G., Andres, S., Slater, L., 2020. Electrical monitoring of saline tracers to reveal subsurface flow pathways in a flat ditch-drained field. J. Hydro. 586, 1–12.
Robinson, D.A., Jones, S.B., Wraith, J.M., Or, D., Friedman, S.P., 2003. A Review of advances in dielectric and electrical conductivity measurement in soils using time domain reflectometry. Vadose Zone J. 2, 444–475.
Robinson, D.A., Lebron, I., Kocar, B., Phan, K., Sampson, M., Crook, N., Fendorf, S., 2009. Time-lapse geophysical imaging of soil moisture dynamics in tropical deltaic soils: an aid to interpreting hydrological and geochemical processes. Water Resour. Res. 45 (4), 1–12.
Romero-Ruiz, A., Linde, N., Keller, T., Or, D., 2018. A review of geophysical methods for soil structure characterization. Rev. Geophys. 56(4), 672–697.
Romkens, M.J.M., Prasad, S.N., 2006. Rain infiltration into swelling/shrinking/cracking soils. Agric. Water Manag. 86 (1–2), 196–205.
Saey, T., De Smedt, P., Islam, M.M., Meerschman, E., Van De Vijver, E., Lehouck, A., Van Meirvenne, M., 2012. Depth slicing of multi-receiver EMI measurements to enhance the delineation of contrasting subsoil features. Geoderma 189–190, 514–521.
Saintenoy, A., Schneider, S., Tucholka, P., 2008. Evaluating ground penetrating radar use for water infiltration monitoring. Vadose Zone J. 7 (1), 208–214.
Sammartino, S., Michel, E., Capowiez, Y., 2012. A novel method to visualize and characterize preferential flow in undisturbed soil cores by using multislice helical CT. Vadose Zone J. 11 (1), 1–13.
Sammartino, S., Lissy, A.-S., Bogner, C., Van Den Bogaert, R., Capowiez, Y., Ruy, S., Cornu, S., 2015. Identifying the functional macropore network related to preferential flow in structured soils. Vadose Zone J. 14 (10), 1–16.
Samouelian, A., Cousin, I., Tabbagh, A., Bruand, A., Richard, G., 2005. Electrical resistivity survey in soil science: a review. Soil Tillage Res. 83, 173–193.
Sass, O., Friedmann, A., Haselwanter, G., Wetzel, K.-F., 2010. Investigating thickness and internal structure of alpine mires using conventional and geophysical techniques. Catena 80 (3), 195–203.
Schmelzbach, C., Tronicke, J., Dietrich, P., 2012. High-resolution water content estimation from surface-based ground-penetrating radar reflection data by impedance inversion. Water Resource Res. 48, 1–16.
Sivapalan, M., 2005. Pattern, process and function: elements of a unified theory of hydrology at the catchment scale. In: Encyclopedia of hydrological sciences. John Wiley & Sons, Ltd., Chichester, UK.
Smettem, K.R.J., Smith, R.E., 2002. Field measurement of infiltration parameters. In: Smith, R.E., Smettem, K.R.J., Broadbridge, P., Woolhiser, D.A. (Eds.) Infiltration theory for hydrologic applications. The American Geophysical Union, Washington DC, U.S.A., pp. 135–157.
Soupios, P.M., Kouli, M., Vallianatos, F., Vafidis, A., Stavroulakis, G., 2007. Estimation of aquifer hydraulic parameters from surficial geophysical methods: a case study of Keritis Basin in Chania (Crete – Greece). J. Hydrol. 338 (1–2), 122–131.
Steelman, C.M., Endres, A.L., Jones, J.P., 2012. High-resolution ground-penetrating radar monitoring of soil moisture dynamics: field results, interpretation, and comparison with unsaturated flow model. Water Resour. Res. 48 (9), 1–17. https://doi.org/10.1029/2011WR011414.
Stephens, D.B., 1995. Vadose zone hydrology. CRC, Lewis Publishers, Boca Raton.
Thompson, D.W., 1942. On growth and form. Cambridge University Press.
Toutin, T., 2011. State-of-the-art of geometric correction of remote sensing data: a data fusion perspective. Int. J. Image Data Fusion 2, 3–35.
Triantafilis, J., Ribeiro, J., Page, D., Santos, F.A.M., 2013. Inferring the location of preferential flow paths of a leachate plume by using a DUALEM-421 and a quasi-three-dimensional inversion model. Vadose Zone J. 12, 1–11.
Truss, S., Grasmueck, M., Vega, S., Viggiano, D.A., 2007. Imaging rainfall drainage within the Miami oolitic limestone using high-resolution time-lapse ground-penetrating radar. Water Resour. Res. 43, 1–15.

van der Kruk, J., Gueting, N., Klotzsche, A., He, G., Rudolph, S., von Hebel, C., Yang, X., Weihermüller, L., Mester, A., Vereecken, H., 2015. Quantitative multi-layer electromagnetic induction inversion and full-waveform inversion of crosshole ground penetrating radar data. J. Earth Sci. 26 (6), 844–850.
Walvoord, M.A., Kurylyk, B.L., 2016. Hydrologic impacts of thawing permafrost-a review. Vadose Zone J. 15 (6), 1–20.
Wu, Y., Guo, L., Cui, X., Chen, J., Cao, X., Lin, H., 2014. Ground-penetrating radar-based automatic reconstruction of three-dimensional coarse root system architecture. Plant Soil 383 (1–2), 155–172.
Xu, X., Li, J., Qiao, X., Fang, G., 2019. Fusion of multiple time-domain GPR datasets of different center frequencies. Near Surf. Geophys. 17, 141–150.
Zenone, T., Morelli, G., Teobaldelli, M., Fischanger, F., Matteucci, M., Sordini, M., Armani, A., Ferrè, C., Chiti, T., Seufert, G., 2008. Preliminary use of ground-penetrating radar and electrical resistivity tomography to study tree roots in pine forests and poplar plantations. Funct. Plant Biol. 35 (10), 1047–1058.
Zhang, S., Hopkins, I., Guo, L., Lin, H., 2019. Dynamics of infiltration rate and field-saturated soil hydraulic conductivity in a wastewater-irrigated cropland. Water 11, 1–21. https://doi.org/10.3390/w11081632.
Zhang, J., Lin, H., Doolittle, J., 2014. Soil layering and preferential flow impacts on seasonal changes of GPR signals in two contrasting soils. Geoderma 213, 560–569.
Zhao, S., Al-Qadi, I., 2017. Pavement drainage pipe condition assessment by GPR image reconstruction using FDTD modeling. Constr. Build. Mater. 154, 1283–1293.
Zhu, Q., Lin, H., Doolittle, J.A., 2010b. Repeated electromagnetic induction surveys for determining subsurface hydrologic dynamics in an agricultural landscape. Soil Sci. Soc. Am. J. 74 (5), 1750–1762.
Zhu, Q., Lin, H., Doolittle, J.A., 2010a. Repeated electromagnetic induction surveys for improved soil mapping in an agricultural landscape. Soil Sci. Soc. Am. J. 74 (5), 1750–1762.